	Design Calculation Sheet	Project no:	IN 18036-0100D
		Date	11/10/2025
		computed by	O.T
	1- Rib 12mm thickness - [+ve σbot. - Hogging] - [P200-Cut]	Approved by	R.AZ

Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490			N/mm^2
span =	2800			mm

Straining actions :

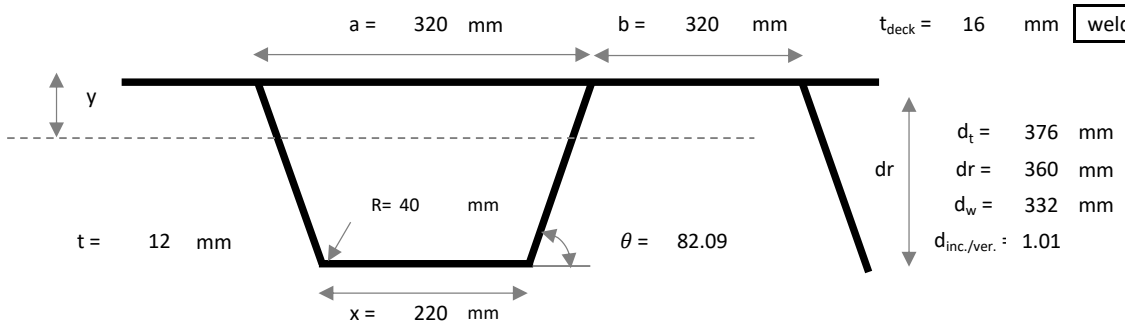
Hogging

M_x =	62.15	KN.m
V_y =	151.3	KN
Axial=	39.93	KN
T =	3.501	KN.m

Refrences

Ult.	0.15 : U.R
Fatigue	0.50 : U.R
weld	0.83 : U.R

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Hogging

Stresses on Gross Section :

σ_{top} = N/A + M/S = 1.898 - 18.81 = -17 MPa compression -ve

$\sigma_{bot.}$ = N/A + M/S = 1.898 + 39.42 = 41 MPa tension +ve

Local buckling of top Flanges : compression -ν - check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\epsilon\sqrt{k})$ = 0.43 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

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dar making progress together		Date	11/10/2025
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k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\varepsilon\sqrt{k}) =$ 0.43 ≤ 0.673

ρ = $(\lambda - 0.055(3+\psi))/\lambda^2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

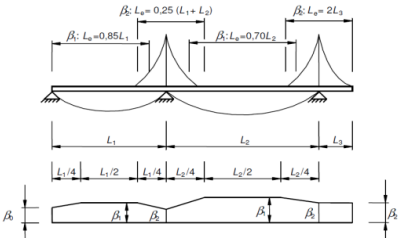
bot. flange : c_3 = 140 mm c_3/t = 11.67 ≤ 42ϵ = 34.42

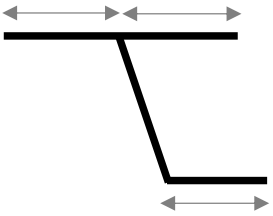
t = 12 mm - No check local buckling

ρ = 1

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Hogging

Le = 0.5 L = 1400 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm k = α bo / Le = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o2} = 160 mm k = α bo / Le = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o3} = 110 mm k = α bo / Le = 0.079 > 0.02

without Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.69 $b_{eff} = B^k b_{c\text{-}eff}$ = 106.8 mm

b_{red} = 3.169 mm

	Design Calculation Sheet	Project no:	IN 18036-0100D
dar making progress together		Date	11/10/2025
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	1- Rib 12mm thickness - [+ve σbot. - Hogging] - [P200-Cut]	Approved by	R.AZ

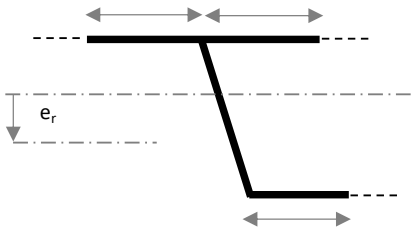
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 9.268 mm b_{red} = 9.268 mm

150.7 mm 150.7 mm



$A_{red\text{-}up}$ = 593.2 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 76.06 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20365.4 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 2.377 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

s_{top} = 3139228 mm^3

s_{bot} = 1541946.7 mm^3

Z= 2319809.0 mm^3 Yu= 16.8 mm

Z_f= 990839.4 mm^3 Yu= 10.13 mm w.o web

b_{red} = 3.169 mm 106.8 mm

t_r = 12 mm

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 335.2 mm t = 12 mm

c/t = 27.93 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

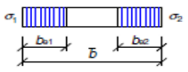
ψ = 1 ≤ 1 k = $8.2/(1.05+\psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\,\sqrt{k}} =$$ 0.6 ≤ 0.673 ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1.00

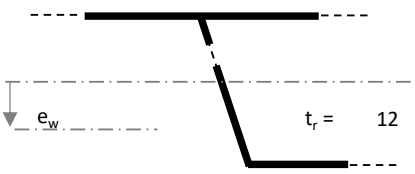
b_c = 335.2 mm b_{eff} = 335.2 mm $y_{red\text{-}top\text{-}web}$= 167.6 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -59.7 mm

A_{new} = 20365.4 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

y_{top} = 123.9 mm s_{top} = 3139228.0 mm^3

$y_{bot.}$ = 252.1 mm s_{bot} = 1541946.7 mm^3

	Design Calculation Sheet	Project no:	IN 18036-0100D
dar making progress together		Date	11/10/2025
		computed by	O.T
	1- Rib 12mm thickness - [+ve σbot. - Hogging] - [P200-Cut]	Approved by	R.AZ

Stresses on Effective Section :

σ_{top} = N/A + M/S = 1.961 - 19.8 = -18 MPa compression -ve

$\sigma_{bot.}$ = N/A + M/S = 1.961 + 40.3 = 42 MPa tension +ve

Final Effective Section Properties for sagging :

A_{new} = 20365 mm2 y_{top} = 123.9 mm S_{top} = 3139228 mm^3

I_x = 388799069 mm^3 $y_{bot.}$ = 252.1 mm S_{bot} = 1541947 mm^3

Check stresses :

IRC 24-2010

Stress due to mx :

clause 510.2

$M_x + N\,e_y$ = 62.15 KN.m $F_{y\text{-actual}}$ = 350 MPa $M_{xdv\text{-top}}$ = 998.8 KN.m OK 0.06 : U.R

$F_{y\text{-actual}}$ = 350 MPa $M_{xdv\text{-bot}}$ = 490.6 KN.m OK 0.13 : U.R

Stress due to Vy :

IRC 24-2010

clause 509.4

d = 363.5 mm t = 12.0 mm $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1301 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.394 ≤ 0.8 τ_b = $F_{yw}/(\sqrt{3})$ 202.1 202.1 MPa

≤ 1.2 $[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 267.7

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1301

Check shear Stress

d/t_w = 27.93 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 no web shear buckling Vn = Vp

OK

V_n = ##### KN $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 151.35 KN 0.09 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 17.35 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T / (2A_o t_{deck})$ = 1.171 MPa < fd x= 183.7 MPa 0.01 : U.R

$fv_{rib} = T / (2A_o t_{rib})$ = 1.562 MPa < fd y= 183.7 MPa

$fv_{rib\text{-total}}$ = 18.91 MPa < fd y= 183.7 MPa 0.10 : U.R

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Axial stress : Axial= 39.93 KN Tension < Pd= 6479.9 KN OK — 0.01 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.26 < 180 OK $\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}}$ = 0.27

Buckling Class b α = 0.34 $\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2]$ = 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right]$ = 0.98 $f_{crx} = \chi f_y / \gamma_{mo}$ = 310.3 MPa

Ae= 20365.4 mm^2 P_d= 6318.4 KN

-Tension : Ae= 20365.4 mm^2 T_d= 6479.9 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{39.93}{6480} - \frac{62.15}{998.8} = 0.01 - 0.06 = 0.06 < 1.0$ OK — 0.06 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{39.93}{6480} + \frac{62.15}{490.6} = 0.01 + 0.13 = 0.13 < 1.0$ OK — 0.13 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} =$ 0.00 + 0.06 = 0.06 < 1.0 OK — clause 510.3.2, 0.06 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$

Top Deck :	$\frac{-17.8^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{1.2^2}{fy/\gamma_{mo}}$	= $\frac{18.0}{318.2}$	= 0.06	≤ 1.00	OK	0.06 : U.R
Top Rib Web :	$\frac{-15.3^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{18.9^2}{fy/\gamma_{mo}}$	= $\frac{36.1}{318.2}$	= 0.11	≤ 1.00	OK	0.11 : U.R
Bott. Rib Web :	$\frac{-36.4^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{18.9^2}{fy/\gamma_{mo}}$	= $\frac{49.0}{318.2}$	= 0.15	≤ 1.00	OK	0.15 : U.R
Bott. Rib Flange	$\frac{-38.3^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{1.6^2}{fy/\gamma_{mo}}$	= $\frac{38.4}{318.2}$	= 0.12	≤ 1.00	OK	0.12 : U.R

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Fatigue : 0.50 : U.R

Input Data :

γ_{mft} = 1.15

γ_{fft} = 1

N_{sc} = 10 $x10^6$ cycle

Straining actions :

M_x (+)= 53.99 KN.m

Axial= 3.66 KN

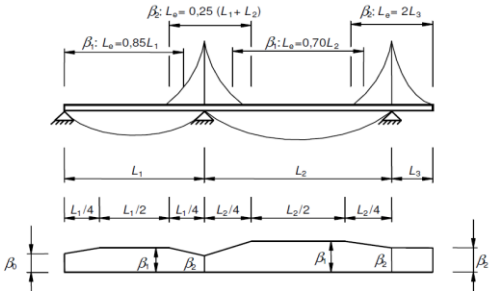
V_y = 84.85 KN

T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

Hogging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :

due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm $b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm

b_{red} = 65.08 mm b_{red} = 65.08 mm

$A_{red\text{-}up}$ = 4165 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 820.3 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 16049.3 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 16.75 mm

t_{deck} = 16 mm

t_r = 12 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 292368927 mm^4

s_{top} = 2115218 mm^3 $y_{red.cg}$ = 138.2 mm

s_{bot} = 1229586.1 mm^3

$b_{eff} = Bb_{c\text{-}eff}$ = 75.82 mm

b_{red} = 34.18 mm

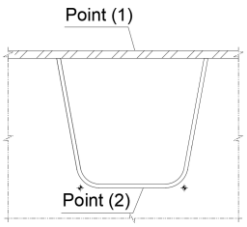
Shear:

A_v = 8723 mm^2

Torsion:

A_o = 93396 mm^2

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	1- Rib 12mm thickness - [+ve σbot. - Hogging] - [P200-Cut]	Approved by	R.AZ



Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 138.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.29 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 237.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 44.14 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.49 : U.R

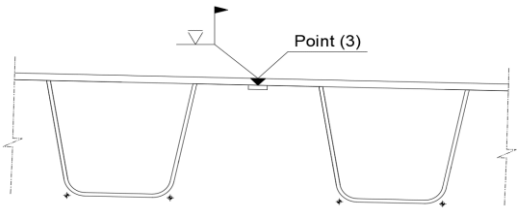
fv x (torsion)= 0.353 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 138.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 72.26 MPa

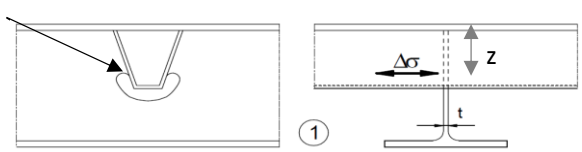
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 62.83 MPa 0.41 : U.R



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Point (4) web at cutout (local effects) Rib web at cutout

z = 230 mm



$\mu_r = \left(\frac{25}{t_f}\right)^{0.25}$ = 1.00 Y = 91.8 mm

f_{fn} = 71 MPa Euro Table (8-8) (1) N=2x10^6

f_{fn} = 52.31 MPa N=5x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 45.54 MPa

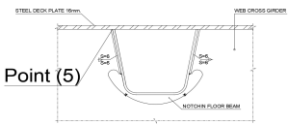
τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 17.18 MPa < $\mu_r f_f / \gamma_{mft}$ = 39.6 MPa 0.43 : U.R

$f_{v\,y}$ = 9.727 MPa $f_{v\,y\,(torsion)}$ = 0.353 MPa

$f_{v\,y(total)}$ = 10.08 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Point (5) web weld (longitudinal stresses) Fillet weld one side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25}$ = 1.00 Y = 122.2 mm

f_{fn} = 66 MPa Table (19-b) iii-(13) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 52.38 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 22.8 MPa < $\mu_r f_f / \gamma_{mft}$ = 45.55 MPa 0.50 : U.R

$f_{v\,y}$ = 9.727 MPa $f_{v\,y\,(torsion)}$ = 0.353 MPa

$f_{v\,y(total)}$ = 10.08 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Design of weld between Rib and Floor beam :

Q= 205 KN Contact length= 275 mm f_{wn} = $f_u/\sqrt{3}$ γmw= 1.25 $f_{wd}=f_{wn}/\gamma mw$ = 226.3 MPa

$\tau_{shear\ flow} = Q\,S_x/I_x$ = 747.03 N/mm' t_{top} = 12 mm a= $\tau/4f$= 1.43 mm

min S_{weld} = 5 mm s= 2.02 mm use S= 6.0 mm 0.83 : U.R

	Design Calculation Sheet	Project no:	IN 18036-0100D
		Date	11/10/2025
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	1- Rib 12mm thickness - [+ve σbot. - Sagging] - [P200-Cut]	Approved by	R.AZ

Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490	N/mm^2		
span =	2800	mm		

Straining actions :

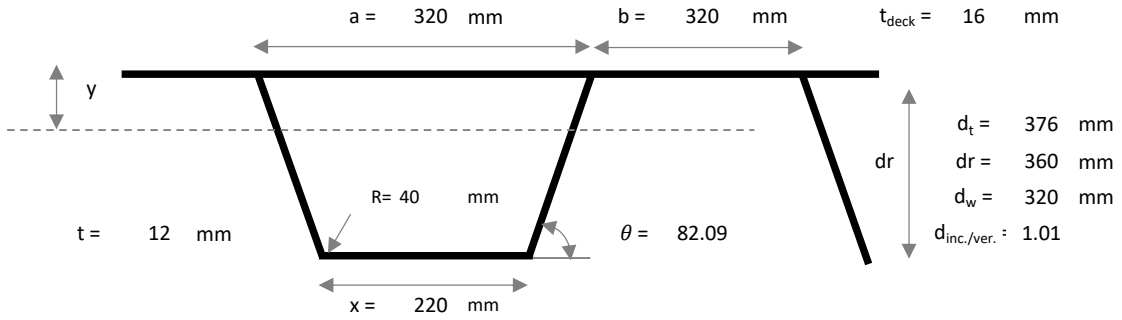
Sagging

M_x =	82.3	KN.m
V_y =	72.9	KN
Axial=	204.5	KN
T =	18.37	KN.m

Refrences

Ult.	0.19 : U.R
Fatigue	0.56 : U.R
Splice	0.99 Plate
	0.84 Bolts

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Sagging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	9.72	-	24.92	=	-15	MPa	compression -ve
$\sigma_{bot.}$ = N/A + M/S =	9.72	+	52.2	=	62	MPa	tension +ve

Local buckling of top Flanges : compression -ν - check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\epsilon\sqrt{k})$ = 0.43 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

	Design Calculation Sheet	Project no:	IN 18036-0100D
dar making progress together		Date	11/10/2025
		computed by	O.T
	1- Rib 12mm thickness - [+ve σbot. - Sagging] - [P200-Cut]	Approved by	R.AZ

k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\varepsilon\sqrt{k}) =$ 0.43 ≤ 0.673

ρ = $(\lambda - 0.055(3+\psi))/\lambda^2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

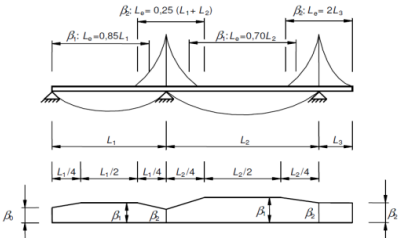
bot. flange : c_3 = 140 mm c_3/t = 11.67 ≤ 42ϵ = 34.42

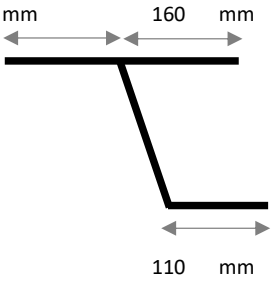
t = 12 mm - No check local buckling

ρ = 1

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Sagging

Le = 0.7 L = 1960 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o2} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o3} = 110 mm k = α bo / Le = 0.056 > 0.02

without Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.98 $b_{eff} = B^k b_{c\text{-}eff}$ = 109.9 mm

b_{red} = 0.123 mm

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		computed by	O.T
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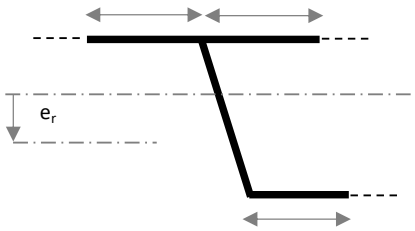
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 0.545 mm b_{red} = 0.545 mm

159.5 mm 159.5 mm



$A_{red\text{-}up}$ = 34.85 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 2.955 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20996.8 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 0.153 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

s_{top} = 3293897 mm^3 y_{top} = 121.6 mm b_{red} = 0.123 mm 109.9 mm

s_{bot} = 1574983.3 mm^3 t_r = 12 mm

Z= 2255736.7 mm^3 Yu= 16.15 mm

Z_f= 1016269.7 mm^3 Yu= 10.07 mm w.o web

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 323.1 mm t = 12 mm

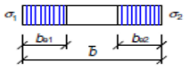
c/t = 26.92 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\,\sqrt{k}} =$$ 0.578 ≤ 0.673 ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1.00

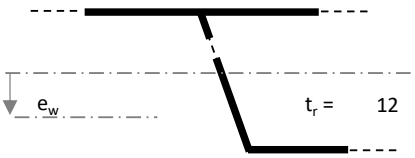
b_c = 323.1 mm b_{eff} = 323.1 mm $y_{red\text{-}top\text{-}web}$= 161.5 mm



b_t = 0 mm b_{red} = 0 mm

Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -55.9 mm

A_{new} = 20996.8 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

y_{top} = 121.6 mm s_{top} = 3293896.7 mm^3

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4 $y_{bot.}$ = 254.4 mm s_{bot} = 1574983.3 mm^3

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		computed by	O.T
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Stresses on Effective Section :

σ_{top} = N/A + M/S = 9.738 - 24.99 = -15 MPa compression -ve

$\sigma_{bot.}$ = N/A + M/S = 9.738 + 52.26 = 62 MPa tension +ve

Final Effective Section Properties for sagging :

A_{new} = 20997 mm^2 y_{top} = 121.6 mm S_{top} = 3293897 mm^3

I_x = 400631140 mm^4 $y_{bot.}$ = 254.4 mm S_{bot} = 1574983 mm^3

Check stresses :

IRC 24-2010 clause 510.2

Stress due to mx :

$M_x + N e_y$ = 82.3 KN.m $F_{y\text{-actual}}$ 350 MPa $M_{xdv\text{-top}}$: 1048 KN.m OK 0.08 : U.R

$F_{y\text{-actual}}$ 350 MPa $M_{xdv\text{-bot}}$: 501.1 KN.m OK 0.16 : U.R

Stress due to Vy :

IRC 24-2010 clause 509.4

d = 363.5 mm t = 12.0 mm $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1401 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.38 ≤ 0.8 τ_b = $F_{yw}/(\sqrt{3})$ 202.1 202.1 MPa

≤ 1.2 $[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 270

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1401

Check shear Stress

d/t_w = 26.92 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 no web shear buckling Vn = Vp

OK

V_n = ##### KN $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 72.903 KN 0.05 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 8.3576 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T / (2A_o t_{deck})$ = 6.146 MPa < fd x= 183.7 MPa 0.03 : U.R

$fv_{rib} = T / (2A_o t_{rib})$ = 8.195 MPa < fd y= 183.7 MPa

$fv_{rib\text{-total}}$ = 16.55 MPa < fd y= 183.7 MPa 0.09 : U.R

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Axial stress : Axial= 204.5 KN Tension < Pd= 6680.8 KN OK — 0.03 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.27 < 180 OK $\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}} =$ 0.27

Buckling Class b α = 0.34 $\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2] =$ 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right] =$ 0.98 $f_{crx} = \chi f_y / \gamma_{mo}$ = 310.2 MPa

Ae= 20996.8 mm^2 P_d= 6514.1 KN

-Tension : Ae= 20996.8 mm^2 T_d= 6680.8 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{204.5}{6681} - \frac{82.3}{1048} =$ 0.03 − 0.08 = 0.05 < 1.0 OK — 0.05 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{204.5}{6681} + \frac{82.3}{501.1} =$ 0.03 + 0.16 = 0.19 < 1.0 OK — 0.19 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} =$ 0.00 + 0.08 = 0.08 < 1.0 OK — clause 510.3.2, 0.08 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$$

Top Deck :	$\frac{-15.2^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{6.1^2}{fy/\gamma_{mo}}$	= $\frac{18.6}{318.2}$	= 0.06	≤ 1.00	OK	0.06 : U.R
Top Rib Web :	$\frac{-12.0^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{16.6^2}{fy/\gamma_{mo}}$	= $\frac{31.1}{318.2}$	= 0.10	≤ 1.00	OK	0.10 : U.R
Bott. Rib Web :	$\frac{-40.1^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{16.6^2}{fy/\gamma_{mo}}$	= $\frac{49.3}{318.2}$	= 0.15	≤ 1.00	OK	0.15 : U.R
Bott. Rib Flange	$\frac{-42.5^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{8.2^2}{fy/\gamma_{mo}}$	= $\frac{44.8}{318.2}$	= 0.14	≤ 1.00	OK	0.14 : U.R

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Fatigue : 0.56 : U.R

Input Data :

γ_{mft} = 1.15
γ_{fft} = 1
N_{sc} = 10 $x10^6$ cycle
Impact Magnification factor : 1.000

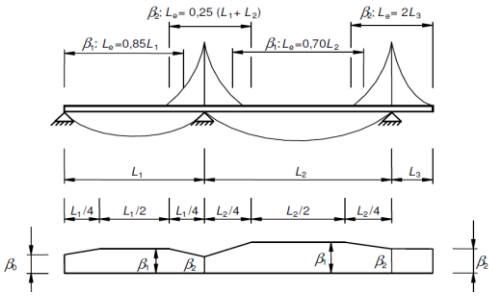
Straining actions :

M_x (+)= 46.57 KN.m
Axial= 2.072 KN
V_y = 81.54 KN
T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

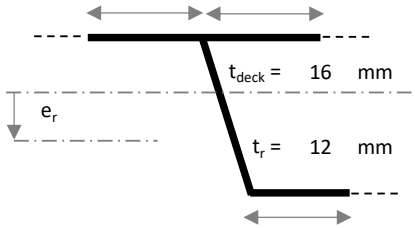
Sagging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :
due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm $b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm
b_{red} = 6.545 mm b_{red} = 6.545 mm

$A_{red\text{-}up}$ = 418.9 mm^2 $y_{red\text{-}olc}$ 113.5 mm
$A_{red\text{-}bot}$ = 52.17 mm^2 $y_{red\text{-}olc}$ 248.5 mm
A_{new} = 20563.6 mm^2
$e_r = (A_{red}\, y_{old})/A_{new}$ = 1.681 mm



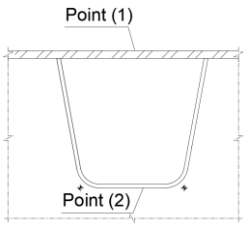
$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 392580543 mm^4

s_{top} = 3187672 mm^3 $y_{red.cg}$ = 123.2 mm $b_{eff} = Bb_{c\text{-}eff}$ = 107.8 mm
s_{bot} = 1552658.3 mm^3 b_{red} = 2.174 mm

Shear:
A_v = 8723 mm^2

Torsion:
A_o = 93396 mm^2

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Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 $Y_{top} =$ 123.2 mm

$f_{fn} =$ 118 MPa Table (19-a) i-(1) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 102.7 MPa

$\tau_{fn} =$ 75 MPa $\tau_f = \tau_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 14.71 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 89.33 MPa 0.16 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r\, \tau_f / \gamma_{mft} =$ 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.} =$ 252.8 mm

$f_{fn} =$ 118 MPa Table (19-a) i-(1) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 102.7 MPa

$\tau_{fn} =$ 75 MPa $\tau_f = \tau_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 30.1 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 89.33 MPa 0.34 : U.R

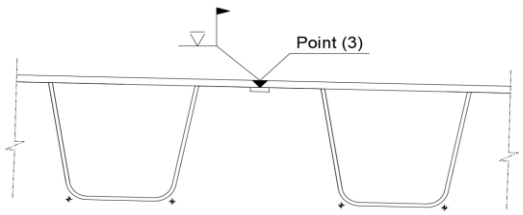
fv x (torsion)= 0.353 MPa < $\mu_r\, \tau_f / \gamma_{mft} =$ 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 123.2 mm

$f_{fn} =$ 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 72.26 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 14.71 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 62.83 MPa 0.23 : U.R



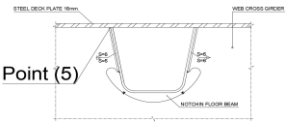
Point (4) web at cutout (local effects) Rib web at cutout

Not applicable

0.00 : U.R

0.00 : U.R

Point (5) web weld (longitudinal stresses) Butt & Fillet weld two side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 107.2 mm

f_{fn} = 92 MPa Table (19-b) i-(9) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 73.02 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 12.81 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 63.5 MPa 0.20 : U.R

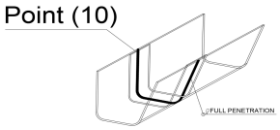
fv y= 9.348 MPa fv y (torsion)= 0.353 MPa

$f_{v\,y(total)}$ = 9.70 MPa < $\mu_r\, \tau_f / \gamma_{mft}$ = 56.7 MPa 0.17 : U.R

Point (10) Welded splice Welded rib splice

Straining actions of Welded splice at location of 0.7m from F.B

Mx =	N
37.4	15
kN.m	kN



$\mu_r = \left(\frac{25}{t_r}\right)^{0.25}$ =1.000 $Y_{bot.}$ = 252.8 mm

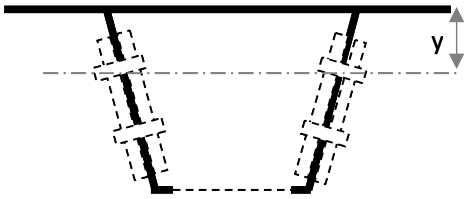
f_{fn} = 80 MPa Euro Table (8-8) (4) N=2x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 51.31 MPa

f_{fn} = 58.94 MPa N=5x10^6

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 24.82 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 44.62 MPa 0.56 : U.R

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Section 2-2 (Splice): Straining action of fatigue at Splice



A =	17637	mm^2	$S_{x\text{-top}}$ =	3031147	mm^3
y =	95	mm	$S_{x\text{-bot}}$ =	1027924	mm^4
I_x =	288621959	mm^4			

web

No.of rows	3
Hole Diameter	30

Point (6) Upper deck Plate Butt weld one side (Transverse)

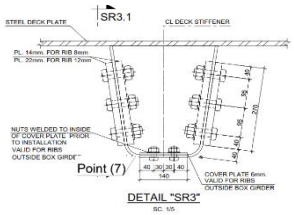
$$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} = 1.000 \qquad Y_{yp.} = 95.2 \text{ mm}$$

f_{fn} = 52 MPa Table (19-b) ix-(23) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 MPa

$$\gamma_{fft}\, f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 15.5 \text{ MPa} \quad < \quad \mu_r\, f_f / \gamma_{mft} = 39.4 \text{ MPa}$$

0.39 : U.R

Point (7) Bottom Flange Plate Net Section (Bollted)

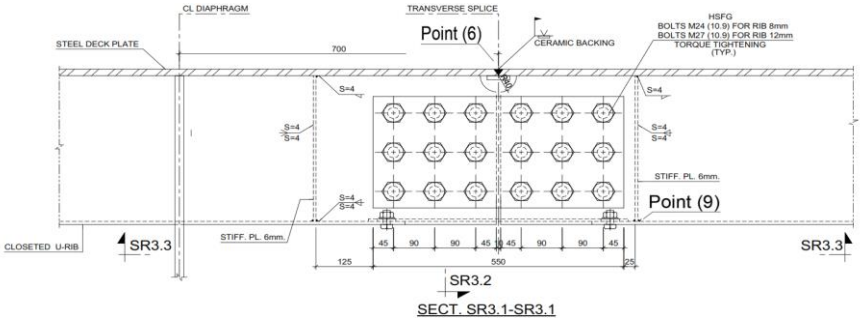


$$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000 \qquad Y_{bot.} = 281 \text{ mm}$$

f_{fn} = 103 MPa Table (19-a) ii-(5) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 89.7 MPa

$$\gamma_{fft}\, f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 45.4 \text{ MPa} \quad < \quad \mu_r\, f_f / \gamma_{mft} = 78 \text{ MPa}$$

0.58 : U.R



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	1- Rib 12mm thickness - [+ve σbot. - Sagging] - [P200-Cut]	Approved by	R.AZ

Point (9) for sagging — Web Vertical stiffeners — Vertical stiffeners Fillet welded

Straining actions and gross section

Mx =	N
0	0

Bolted splice at location of 0.7m and closure plate at 1.0m from F.B

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.} = 240.8$ mm

$f_{fn} =$ 59 MPa Table (19-b) xviii-(36&37) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 51.4 MPa

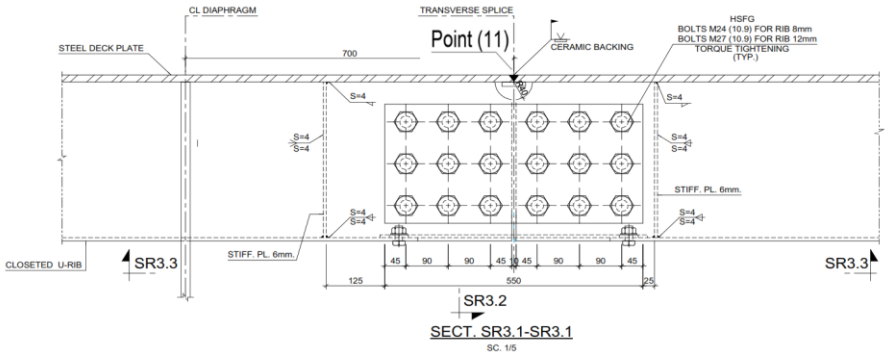
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 0 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 44.7 MPa — 0.00 : U.R

Point (11) — Intermittient Long. Welc — Zones containing cope holes

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.} = 79.2$ mm

$f_{fn} =$ 52 Mpa Table (19-b) v-(15) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 Mpa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 12.9 Mpa < $\mu_r\, f_f / \gamma_{mft} =$ 39.4 Mpa — 0.33 : U.R



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Plate.G : 0.99 : U.R

Bolts : 0.84 : U.R

Splice Straining actions :

M = 82.3 KN.m

Q = 72.9 KN

P = 204.5 KN

T = 18.37 KN.m

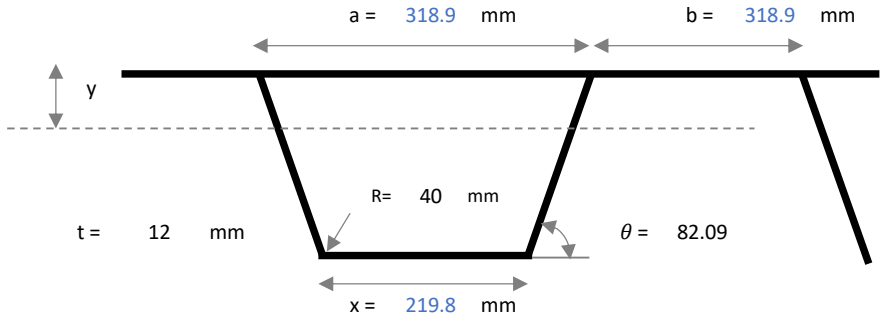
AVERAGE Global Rib stress = -55.26 Mpa

Eq. Global Normal Force = -1160 kN

$N_{Rib\text{-}total}$ = 1365 KN

$F_{y\text{-}actual}$ 350 Mpa

Section Properties :



A = 20997 mm^2

y = 122 mm

I_x = 400631140 mm^4

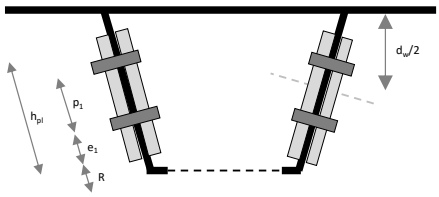
$S_{x\text{-}top}$ = 3293897 mm^3

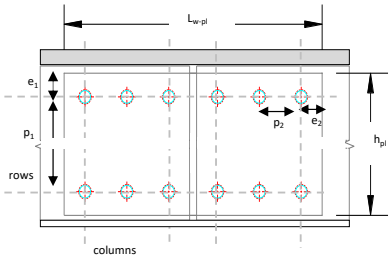
$S_{x\text{-}bot}$ = 1574983 mm^4

$I_{x\text{-}rib}$ = 268650721 mm^4 0.67

Splice plates Dimintions & BOLTES Confiugrations :

Bolts Grade : 10.9





web :

Plates : h_{pl} = 270 mm

< 271 mm

$L_{w\text{-}pl}$ = 532 mm

t_{pl} = 24 mm > 20 mm

> 40 mm

$y_{splice\text{-}web}$ = -13.34 mm $F_{y\ pl.}$ = 330 Mpa

Bolts : M27

No.of Rows 3 $y_{bot\text{-}pl.edge}$ = 43 mm

No.of column 3 $b_{opening}$ 100 mm

e_1 = 41 mm ≥ 1.5 Ø = 40.5 mm

p_1 = 94 mm ≥ 3.0 Ø = 81 mm

e_2 = 40.5 mm ≥ 1.5 Ø = 40.5mm

p_2 = 90 mm ≥ 3.0 Ø = 81 mm

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tension
compression

Check on base material :

(Befor opening net section) :

						$y_{bot\text{-}pl.edge}$	$y_{Bolt\text{-}1}$	$y_{Bolt\text{-}2}$	$y_{Bolt\text{-}3}$
A =	18837	mm^2	$s_{x\text{-}top}$ =	3313146	mm^3	43	84	177	270
y =	113	mm	$s_{x\text{-}bot}$ =	1418548	mm^4				
I_x =	373470022	mm^4							

-Top stresses: (compression)

$$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=\frac{1365}{19}-\frac{82.3}{3.313}=48-17=31.7 < .9f_u/1.25 = 352.8 \quad \text{OK}$$

U.R : 0.09

-Bottom stresses: (tension)

$$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=\frac{1365}{19}-\frac{82.3}{1.419}=48+39=87.0 < .9f_u/1.25 = 352.8 \quad \text{OK}$$

U.R : 0.247

$0.9\,A_n F_u/\,1.25$ =	6646	KN	<	$A_g F_y/\,1.1$ =	6681 KN
$0.9\,Sx_n F_u/\,1.25$ =	500	KN.m	<	$Sx_g F_y/\,1.1$ =	501 KN.m

(After one bolt column at opening section without bolt holes) :

With 0.6667 Force clause-509.6.3.3

						$b_{opening}$
A_{total} =	19797	mm^2	0.94	$s_{x\text{-}top}$ =	3022371	mm^3	100
y =	107	mm		$s_{x\text{-}bot}$ =	1195513	mm^4	
Ix =	322103396	mm^4		t_{web} =	12	mm	

clause 609.6.1.2

-Top stresses: (compression)

$$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=\frac{1365}{20}-\frac{82.3}{3.022}=46-18=27.8 < f_y/1.1 = 318.2 \quad \text{OK}$$

U.R : 0.087

-Bottom stresses: (tension)

$$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=\frac{1365}{20}-\frac{82.3}{1.196}=46+46=91.8 < f_y/1.1 = 318.2 \quad \text{OK}$$

U.R : 0.289

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(After one bolt column at opening net section with bolt holes) : With 0.6667 Force

A_{total} = 17637 mm^2 $S_{x\text{-}top}$ = 3031147 mm^3 clause-509.6.3.3

y = 95 mm $S_{x\text{-}bot}$ = 1027924 mm^4

Ix = 288621959 mm^4

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1365}{18}\ \frac{82.3}{3.031}$ 52 − 18 = 33.5 < .9f_u/1.25 352.8 OK U.R : 0.095

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1365}{18}\ \frac{82.3}{1.028}$ 52 + 53 = 105.0 < .9f_u/1.25 352.8 OK U.R : 0.298

WEB :

Actual Force = 144.29 KN 0.13 : U.R

Yielding	<	V_d = Awg Fy /1.1√3 =	1410.8	KN	clause-506.1.2
Shear Rupture	<	V_{dr} = 0.9 Awn Fu /1.25√3 =	1124.4	KN	
Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	4380.7	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	5126.3	KN	

A_{vn} = 8634.1 mm^2 A_{tn} = 6984 mm^2

A_{vg} = 10434 mm^2 A_{tg} = 10584 mm^2

Actual tension Force A_{rib}/A_{tot} = 675.33 KN 0.24 : U.R

Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	2917.0	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	2776.2	KN	

for Rib : A_{vn} = 6984 mm^2 A_{tn} = 3072 mm^2 t= 12 mm

A_{vg} = 10584 mm^2 A_{tg} = 4512 mm^2

for Splice Plate : A_{vn} = 17568 mm^2 A_{tn} = 12288 mm^2 $2xt_{pl}$ = 48 mm

A_{vg} = 21168 mm^2 A_{tg} = 18048 mm^2

Elliptical BSR :

$\left(\frac{V}{\phi R_{bsv}}\right)^2+\left(\frac{H}{\phi R_{bsh}}\right)^2$ = 0.0329 + 0.2433 = 0.27 < 1 0.27 : U.R

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Check on Splice plates :

rib :

Area web splice Agross = 25920 mm^2 ≥ 1.05 Area rib = 10910.30314 mm^2 clause-509.6.3.3 — 0.42 : U.R

Section :

Ix splice = 422800931.1 mm^4 ≥ 1.05 Ix Section = 420662697 mm^2 — 0.99 : U.R

Check on Splice Bolts :

WEB :

$M_{x\text{-web}}$ =	27.594	KN.m	No.of Rows	3	p1 =	94	mm
$M_{due\ to\ shear}$ =	3.3121	KN.m	No.of column	3	p2 =	90	mm
$N_{one\text{-}web}$ =	337.67	KN	no.of Bolts =	9	$X_{C.G}$ =	90	mm
$V_{Torsion} = T/b_{flange}$ =	71.382	KN			$Y_{C.G}$ =	94	mm
$V_{shear} + V_{Torsion}$ =	107.83	KN					
0.5 $M_{x\text{-capacity-rib}}$ =	168.02	KN.m					

$Q_{v\text{-}y}$ =	11.982	KN	$Q_{T\text{-}y}$ =	27.373	KN	Q_y =	39.355	KN
$Q_{N\text{-}x}$ =	37.518	KN	$Q_{T\text{-}x}$ =	28.59	KN	Q_x =	66.108	KN

1.05 Q_{total} = 80.783 KN one bolt < $V_{nsf}/\gamma = \mu$ n K F/γ = 257.04 KN μ = 0.5 — v= 80.01

0.5 capacity-rib = 215.18 KN < — t= 11.11

$F_o = A_{nb} f_o$ = 321.3 KN n = 2

f_o = 700 N/mm^2 K_h = 1

A_{nb} = 459 mm^2 γ = 1.25 — 0.84 : U.R

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Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490			N/mm^2
span =	2800			mm

Straining actions :

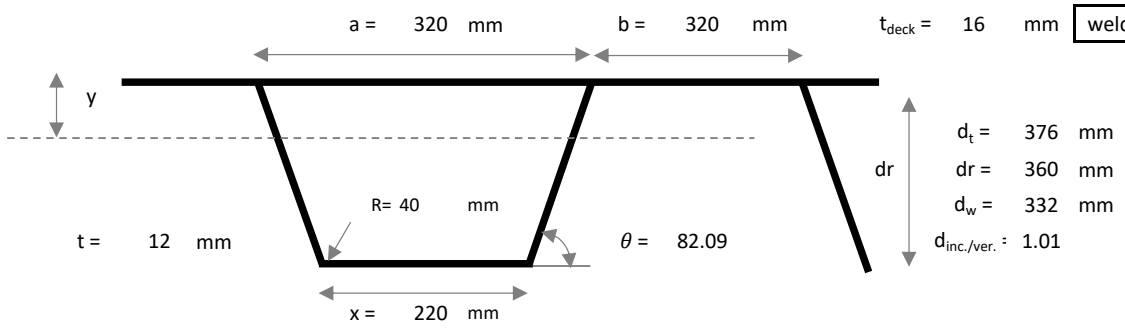
Hogging

M_x =	-137	KN.m
V_y =	184.3	KN
Axial=	-9.57	KN
T =	4.967	KN.m

Refrences

Ult.	0.29 : U.R
Fatigue	0.50 : U.R
weld	0.83 : U.R

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Hogging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	-0.45	-	-41.4	=	41	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-0.45	+	-86.8	=	-87	MPa	compression -ve

Local buckling of top Flanges : tension +ve - No check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-}actual}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

bot. flange : c_3 = 140 mm c_3/t = 11.67 ≤ 42∈ = 34.42

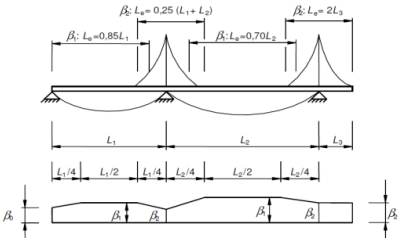
t = 12 mm - No local buckling

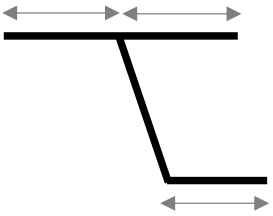
k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\varepsilon\sqrt{k})$ = 0.251 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Hogging

Le = 0.5 L = 1400 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm $k = \alpha\ bo / Le$ = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o2} = 160 mm $k = \alpha\ bo / Le$ = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o3} = 110 mm $k = \alpha\ bo / Le$ = 0.079 > 0.02

without Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.69 $b_{eff} = B^k b_{c\text{-}eff}$ = 106.8 mm

b_{red} = 3.169 mm

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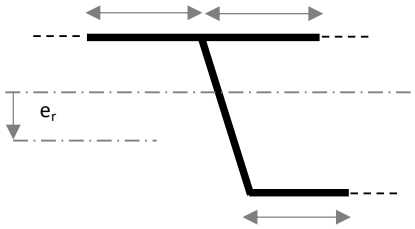
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 9.268 mm b_{red} = 9.268 mm

150.7 mm 150.7 mm



$A_{red\text{-}up}$ = 593.2 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 76.06 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20365.4 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 2.377 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

s_{top} = 3139228 mm^3

s_{bot} = 1541946.7 mm^3

Z= 2319809.0 mm^3 Yu= 16.8 mm

Z_f= 990839.4 mm^3 Yu= 10.13 mm w.o web

b_{red} = 3.169 mm 106.8 mm

t_r = 12 mm

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 335.2 mm t = 12 mm

c/t = 27.93 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

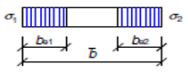
ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\sqrt{k}} =$$ 0.6 ≤ 0.673 ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1.00

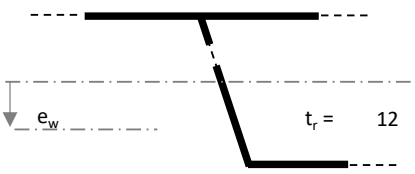
b_c = 335.2 mm b_{eff} = 335.2 mm $y_{red\text{-}top\text{-}web}$= 167.6 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -59.7 mm

A_{new} = 20365.4 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

y_{top} = 123.9 mm s_{top} = 3139228.0 mm^3

$y_{bot.}$ = 252.1 mm s_{bot} = 1541946.7 mm^3

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Stresses on Effective Section :

σ_{top} = N/A + M/S =	-0.47	-	-43.6	=	43	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-0.47	+	-88.8	=	-89	MPa	compression -ve

Final Effective Section Properties for sagging :

A_{new} =	20365	mm2	y_{top} =	123.9	mm	S_{top} =	3139228 mm^3
I_x =	388799069	mm^3	$y_{bot.}$ =	252.1	mm	S_{bot} =	1541947 mm^3

Check stresses :

IRC 24-2010 clause 510.2

Stress due to mx :

$M_x + N e_y$ = -137 KN.m

$F_{y\text{-actual}}$ = 350 MPa — $M_{xdv\text{-top}}$ = 998.8 KN.m OK — 0.14 : U.R

$F_{y\text{-actual}}$ = 350 MPa — $M_{xdv\text{-bot}}$ = 490.6 KN.m OK — 0.28 : U.R

Stress due to Vy :

IRC 24-2010 clause 509.4

d = 363.5 mm t = 12.0 mm $A_v = 2 d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1301 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.394 ≤ 0.8 τ_b = $F_{yw}/(\sqrt{3})$ 202.1 202.1 MPa

≤ 1.2 $[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 267.7

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1301

Check shear Stress

d/t_w = 27.93 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 no web shear buckling Vn = Vp

OK

V_n = ##### KN $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 184.32 KN — 0.12 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 21.131 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T / (2A_o t_{deck})$ =	1.662	MPa	<	fd x=	183.7	MPa	0.01 : U.R
$fv_{rib} = T / (2A_o t_{rib})$ =	2.216	MPa	<	fd y=	183.7	MPa	
$fv_{rib\text{-}total}$ =	23.35	MPa	<	fd y=	183.7	MPa	0.13 : U.R

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Axial stress : Axial= -9.57 KN comp. < Pd= 6318.4 KN OK — 0.00 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.26 < 180 OK $\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}} =$ 0.27

Buckling Class b α = 0.34 $\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2] =$ 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right] =$ 0.98 $f_{crx} = \chi f_y / \gamma_{mo}$ = 310.3 MPa

Ae= 20365.4 mm^2 P_d= 6318.4 KN

-Tension : Ae= 20365.4 mm^2 T_d= 6479.9 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} =$ $\frac{-9.57}{6318} - \frac{-137}{998.8} =$ 0.00 − -0.14 = 0.14 < 1.0 OK — 0.14 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} =$ $\frac{-9.57}{6318} + \frac{-137}{490.6} =$ 0.00 + -0.28 = 0.28 < 1.0 OK — 0.28 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} =$ 0.00 + -0.28 = -0.28 < 1.0 OK — clause 510.3.2 — -0.28 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$$

Top Deck :	$\frac{43.1^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{1.7^2}{fy/\gamma_{mo}}$	=	$\frac{43.2}{318.2}$	= 0.14	≤ 1.00	OK — 0.14 : U.R
Top Rib Web :	$\frac{37.5^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{23.3^2}{fy/\gamma_{mo}}$	=	$\frac{55.1}{318.2}$	= 0.17	≤ 1.00	OK — 0.17 : U.R
Bott. Rib Web :	$\frac{84.1^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{23.3^2}{fy/\gamma_{mo}}$	=	$\frac{93.3}{318.2}$	= 0.29	≤ 1.00	OK — 0.29 : U.R
Bott. Rib Flange	$\frac{88.3^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.2^2}{fy/\gamma_{mo}}$	=	$\frac{88.4}{318.2}$	= 0.28	≤ 1.00	OK — 0.28 : U.R

	Design Calculation Sheet	Project no:	IN 18036-0100D
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		computed by	O.T
	1- Rib 12mm thickness - [+ve σtop - Hogging] - [P200-Cut]	Approved by	R.AZ

Fatigue : 0.50 : U.R

Input Data :

γ_{mft} = 1.15

γ_{fft} = 1

N_{sc} = 10 $x10^6$ cycle

Straining actions :

M_x (+)= 53.99 KN.m

Axial= 3.66 KN

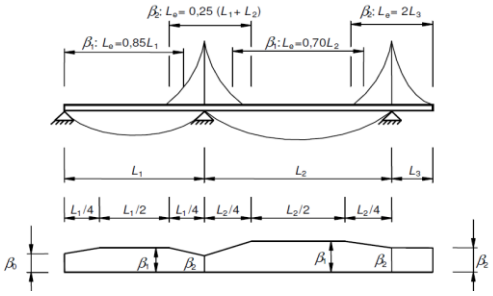
V_y = 84.85 KN

T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

Hogging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :

due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm $b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm

b_{red} = 65.08 mm b_{red} = 65.08 mm

$A_{red\text{-}up}$ = 4165 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 820.3 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 16049.3 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 16.75 mm

t_{deck} = 16 mm

t_r = 12 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 292368927 mm^4

s_{top} = 2115218 mm^3 $y_{red.cg}$ = 138.2 mm

s_{bot} = 1229586.1 mm^3

$b_{eff} = Bb_{c\text{-}eff}$ = 75.82 mm

b_{red} = 34.18 mm

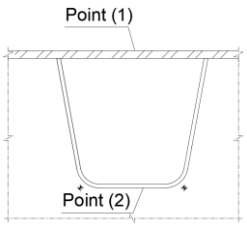
Shear:

A_v = 8723 mm^2

Torsion:

A_o = 93396 mm^2

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	1- Rib 12mm thickness - [+ve σtop - Hogging] - [P200-Cut]	Approved by	R.AZ



Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 138.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.29 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 237.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 44.14 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.49 : U.R

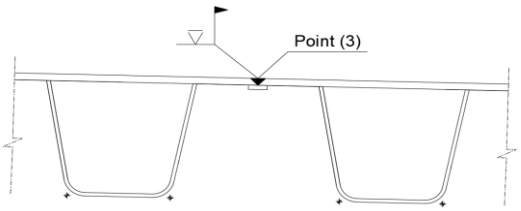
fv x (torsion)= 0.353 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 138.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 72.26 MPa

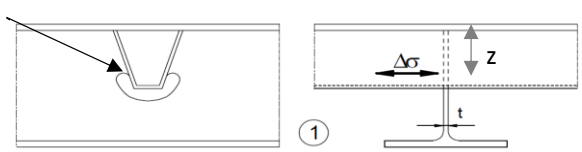
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 62.83 MPa 0.41 : U.R



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Point (4) web at cutout (local effects) Rib web at cutout

z = 230 mm



$\mu_r = \left(\frac{25}{t_f}\right)^{0.25}$ = 1.00 Y = 91.8 mm

f_{fn} = 71 MPa Euro Table (8-8) (1) N=2x10^6

f_{fn} = 52.31 MPa N=5x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 45.54 MPa

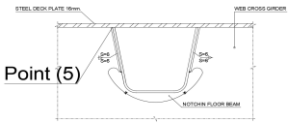
τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 17.18 MPa < $\mu_r f_f / \gamma_{mft}$ = 39.6 MPa 0.43 : U.R

$f_{v\,y}$ = 9.727 MPa $f_{v\,y\,(torsion)}$ = 0.353 MPa

$f_{v\,(total)}$ = 10.08 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Point (5) web weld (longitudinal stresses) Fillet weld one side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25}$ = 1.00 Y = 122.2 mm

f_{fn} = 66 MPa Table (19-b) iii-(13) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 52.38 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 22.8 MPa < $\mu_r f_f / \gamma_{mft}$ = 45.55 MPa 0.50 : U.R

$f_{v\,y}$ = 9.727 MPa $f_{v\,y\,(torsion)}$ = 0.353 MPa

$f_{v\,(total)}$ = 10.08 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Design of weld between Rib and Floor beam :

Q= 205 KN Contact length= 275 mm f_{wn} = $f_u/\sqrt{3}$ γmw= 1.25 $f_{wd}=f_{wn}/\gamma mw$ = 226.3 MPa

$\tau_{shear\ flow} = Q\,S_x/I_x$ = 747.03 N/mm' t_{top} = 12 mm a= $\tau/4f$= 1.43 mm

min S_{weld} = 5 mm s= 2.02 mm use S= 6.0 mm 0.83 : U.R

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	1- Rib 12mm thickness - [+ve σtop - Sagging] - [P200-Cut]	Approved by	R.AZ

Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490	N/mm^2		
span =	2800	mm		

Straining actions :

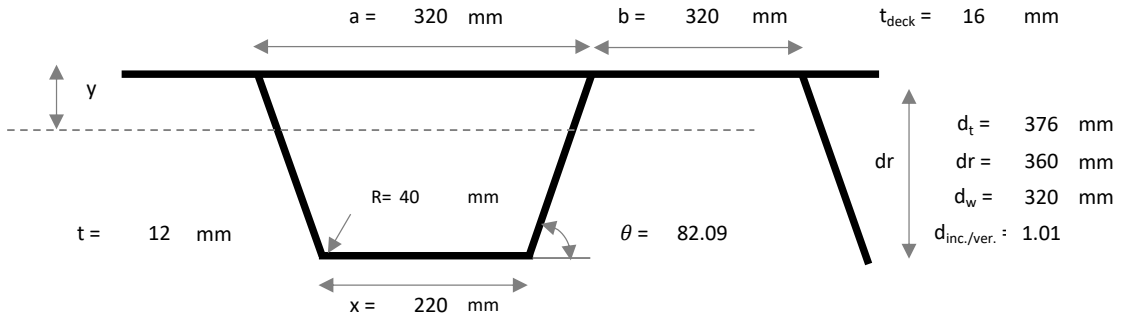
Sagging

M_x =	-113	KN.m
V_y =	153.8	KN
Axial=	-108	KN
T =	6.053	KN.m

Refrences

Ult.	0.24 : U.R
Fatigue	0.56 : U.R
Splice	0.99 Plate
	0.84 Bolts

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Sagging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	-5.12	-	-34.3	=	29	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-5.12	+	-71.9	=	-77	MPa	compression -ve

Local buckling of top Flanges : tension +ve - No check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

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	1- Rib 12mm thickness - [+ve σtop - Sagging] - [P200-Cut]	Approved by	R.AZ

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-}actual}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

bot. flange : c_3 = 140 mm c_3/t= 11.67 ≤ 42∈ = 34.42

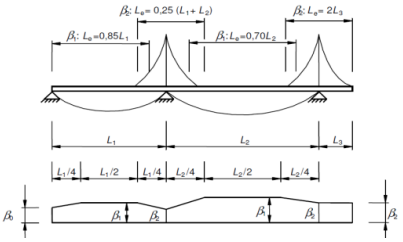
t = 12 mm - No local buckling

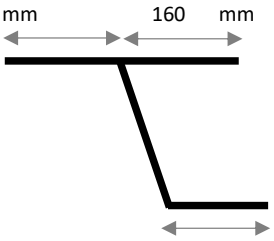
k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\varepsilon\sqrt{k})$= 0.251 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2$= 1 ψ = 1 ≥ -3

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Sagging

Le = 0.7 L = 1960 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o2} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o3} = 110 mm k = α bo / Le = 0.056 > 0.02

without Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.98 $b_{eff} = B^k b_{c\text{-}eff}$ = 109.9 mm

b_{red} = 0.123 mm

	Design Calculation Sheet	Project no:	IN 18036-0100D
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	1- Rib 12mm thickness - [+ve σtop - Sagging] - [P200-Cut]	Approved by	R.AZ

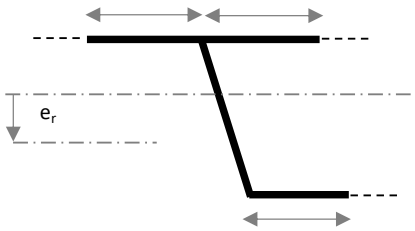
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 0.545 mm b_{red} = 0.545 mm

159.5 mm 159.5 mm



$A_{red\text{-}up}$ = 34.85 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 2.955 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20996.8 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 0.153 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

s_{top} = 3293897 mm^3 y_{top} = 121.6 mm b_{red} = 0.123 mm 109.9 mm

s_{bot} = 1574983.3 mm^3 t_r = 12 mm

Z= 2255736.7 mm^3 Yu= 16.15 mm

Z_f= 1016269.7 mm^3 Yu= 10.07 mm w.o web

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 323.1 mm t = 12 mm

c/t = 26.92 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

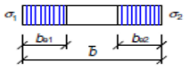
ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\,\sqrt{k}} = 0.578 \le 0.673 \qquad \rho = (\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2 = 1.00$$

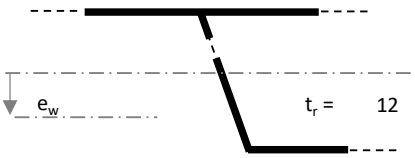
b_c = 323.1 mm b_{eff} = 323.1 mm $y_{red\text{-}top\text{-}web}$ = 161.5 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -55.9 mm

A_{new} = 20996.8 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

y_{top} = 121.6 mm s_{top} = 3293896.7 mm^3

$y_{bot.}$ = 254.4 mm s_{bot} = 1574983.3 mm^3

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	1- Rib 12mm thickness - [+ve σtop - Sagging] - [P200-Cut]	Approved by	R.AZ

Stresses on Effective Section :

σ_{top} = N/A + M/S =	-5.13	-	-34.4	=	29	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-5.13	+	-71.9	=	-77	MPa	compression -ve

Final Effective Section Properties for sagging :

A_{new} =	20997	mm^2	y_{top} =	121.6	mm	S_{top} =	3293897	mm^3
I_x =	400631140	mm^4	$y_{bot.}$ =	254.4	mm	S_{bot} =	1574983	mm^3

Check stresses :

IRC 24-2010 clause 510.2

Stress due to mx :

$M_x + N e_y$ =	-113	KN.m	$F_{y\text{-actual}}$	350	MPa	$M_{xdv\text{-top}}$:	1048	KN.m OK	0.11 : U.R
			$F_{y\text{-actual}}$	350	MPa	$M_{xdv\text{-bot}}$:	501.1	KN.m OK	0.23 : U.R

Stress due to Vy :

IRC 24-2010 clause 509.4

d = 363.5 mm t = 12.0 mm $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1401 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.38 ≤ 0.8 τ_b = $F_{yw}/(\sqrt{3})$ 202.1 202.1 MPa

≤ 1.2 [1-0.8(λw-0.8)](Fy/√3)= 270

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$= 1401

Check shear Stress

d/t_w = 26.92 ≤ 67 ε√(K_v/5.35) = 54.9 no web shear buckling Vn = Vp

OK

V_n= ##### KN $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 153.77 KN 0.10 : U.R

$f_{d\,y}$= 183.7 MPa > $f_{v\,y}$= 17.629 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T / (2A_o t_{deck})$ =	2.025	MPa	<	fd x=	183.7	MPa	0.01 : U.R
$fv_{rib} = T / (2A_o t_{rib})$ =	2.701	MPa	<	fd y=	183.7	MPa	
$fv_{rib\text{-total}}$=	20.33	MPa	<	fd y=	183.7	MPa	0.11 : U.R

dar making progress together	Design Calculation Sheet	Project no:	IN 18036-0100D
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	1- Rib 12mm thickness - [+ve σtop - Sagging] - [P200-Cut]	Approved by	R.AZ

Axial stress : Axial= -108 KN comp. < Pd= 6514.1 KN OK — 0.02 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.27 < 180 OK $\quad \lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}} =$ 0.27

Buckling Class b α = 0.34 $\quad \emptyset = 0.5[1+\alpha(\lambda-0.2)+\lambda^2] =$ 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right] =$ 0.98 $\quad f_{crx} = \chi f_y / \gamma_{mo} =$ 310.2 MPa

Ae= 20996.8 mm^2 $\quad P_d =$ 6514.1 KN

-Tension : Ae= 20996.8 mm^2 $\quad T_d =$ 6680.8 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{-108}{6514} - \frac{-113}{1048} =$ -0.02 − -0.11 = 0.09 < 1.0 OK — 0.09 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{-108}{6514} + \frac{-113}{501.1} =$ -0.02 + -0.23 = 0.24 < 1.0 OK — 0.24 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} =$ 0.02 + -0.23 = -0.21 < 1.0 OK — clause 510.3.2, -0.21 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$$

Top Deck :	$\frac{29.3^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.0^2}{fy/\gamma_{mo}}$	=	$\frac{29.5}{318.2}$	= 0.09	≤ 1.00	OK — 0.09 : U.R
Top Rib Web :	$\frac{24.7^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{20.3^2}{fy/\gamma_{mo}}$	=	$\frac{43.0}{318.2}$	= 0.14	≤ 1.00	OK — 0.14 : U.R
Bott. Rib Web :	$\frac{63.4^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{20.3^2}{fy/\gamma_{mo}}$	=	$\frac{72.5}{318.2}$	= 0.23	≤ 1.00	OK — 0.23 : U.R
Bott. Rib Flange	$\frac{66.8^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.7^2}{fy/\gamma_{mo}}$	=	$\frac{67.0}{318.2}$	= 0.21	≤ 1.00	OK — 0.21 : U.R

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dar making progress together		Date	11/10/2025
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Fatigue : 0.56 : U.R

Input Data :

γ_{mft} = 1.15
γ_{fft} = 1
N_{sc} = 10 $x10^6$ cycle
Impact Magnification factor : 1.000

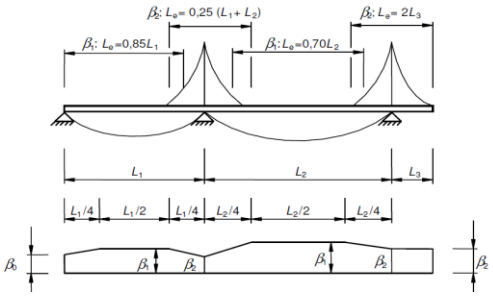
Straining actions :

M_x (+)= 46.57 KN.m
Axial= 2.072 KN
V_y = 81.54 KN
T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

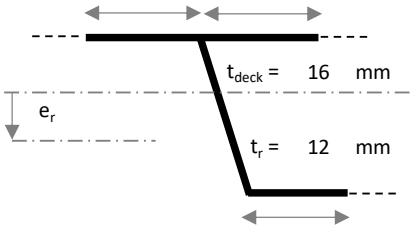
Sagging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :
due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm $b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm
b_{red} = 6.545 mm b_{red} = 6.545 mm

$A_{red\text{-}up}$ = 418.9 mm^2 $y_{red\text{-}olc}$ 113.5 mm
$A_{red\text{-}bot}$ = 52.17 mm^2 $y_{red\text{-}olc}$ 248.5 mm
A_{new} = 20563.6 mm^2
$e_r = (A_{red}\, y_{old})/A_{new}$ = 1.681 mm



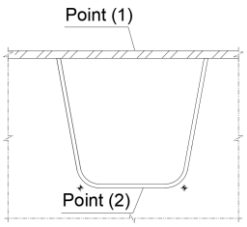
$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 392580543 mm^4

s_{top} = 3187672 mm^3 $y_{red.cg}$ = 123.2 mm $b_{eff} = Bb_{c\text{-}eff}$ = 107.8 mm
s_{bot} = 1552658.3 mm^3 b_{red} = 2.174 mm

Shear:
A_v = 8723 mm^2

Torsion:
A_o = 93396 mm^2

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Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 123.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 14.71 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.16 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 252.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 30.1 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.34 : U.R

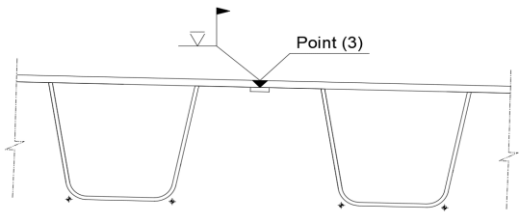
fv x (torsion)= 0.353 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 123.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 72.26 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 14.71 MPa < $\mu_r f_f / \gamma_{mft}$ = 62.83 MPa 0.23 : U.R



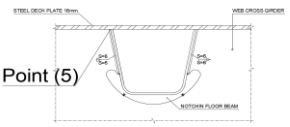
Point (4) web at cutout (local effects) Rib web at cutout

Not applicable

0.00 : U.R

0.00 : U.R

Point (5) web weld (longitudinal stresses) Butt & Fillet weld two side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 107.2 mm

f_{fn} = 92 MPa Table (19-b) i-(9) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 73.02 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 12.81 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 63.5 MPa 0.20 : U.R

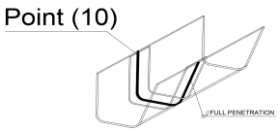
fv y= 9.348 MPa fv y (torsion)= 0.353 MPa

$f_{v\,y(total)}$ = 9.70 MPa < $\mu_r\, \tau_f / \gamma_{mft}$ = 56.7 MPa 0.17 : U.R

Point (10) Welded splice Welded rib splice

Straining actions of Welded splice at location of 0.7m from F.B

Mx =	N
37.4	15
kN.m	kN



$\mu_r = \left(\frac{25}{t_r}\right)^{0.25}$ =1.000 $Y_{bot.}$ = 252.8 mm

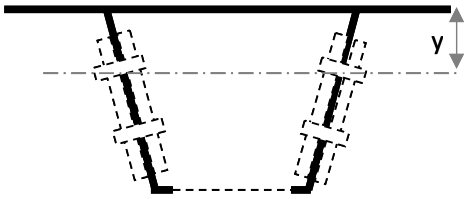
f_{fn} = 80 MPa Euro Table (8-8) (4) N=2x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 51.31 MPa

f_{fn} = 58.94 MPa N=5x10^6

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 24.82 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 44.62 MPa 0.56 : U.R

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Section 2-2 (Splice): Straining action of fatigue at Splice



A =	17637	mm^2	$S_{x\text{-top}}$ =	3031147	mm^3
y =	95	mm	$S_{x\text{-bot}}$ =	1027924	mm^4
I_x =	288621959	mm^4			

web

No.of rows	3
Hole Diameter	30

Point (6) Upper deck Plate Butt weld one side (Transverse)

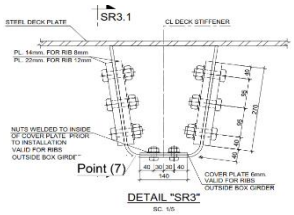
$$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} = 1.000 \qquad Y_{yp.} = 95.2 \text{ mm}$$

f_{fn} = 52 MPa Table (19-b) ix-(23) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 MPa

$$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 15.5 \text{ MPa} \quad < \quad \mu_r f_f / \gamma_{mft} = 39.4 \text{ MPa}$$

0.39 : U.R

Point (7) Bottom Flange Plate Net Section (Bollted)

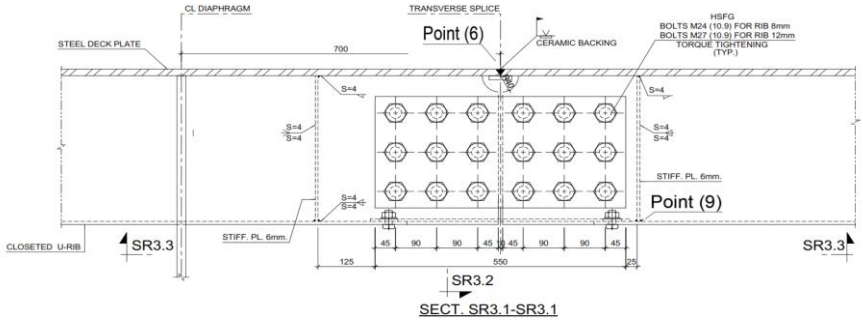


$$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000 \qquad Y_{bot.} = 281 \text{ mm}$$

f_{fn} = 103 MPa Table (19-a) ii-(5) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 89.7 MPa

$$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 45.4 \text{ MPa} \quad < \quad \mu_r f_f / \gamma_{mft} = 78 \text{ MPa}$$

0.58 : U.R



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Point (9) for sagging — Web Vertical stiffeners — Vertical stiffeners Fillet welded

Straining actions and gross section

Mx =	N
0	0

Bolted splice at location of 0.7m and closure plate at 1.0m from F.B

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.}$ = 240.8 mm

f_{fn} = 59 MPa Table (19-b) xviii-(36&37) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 51.4 MPa

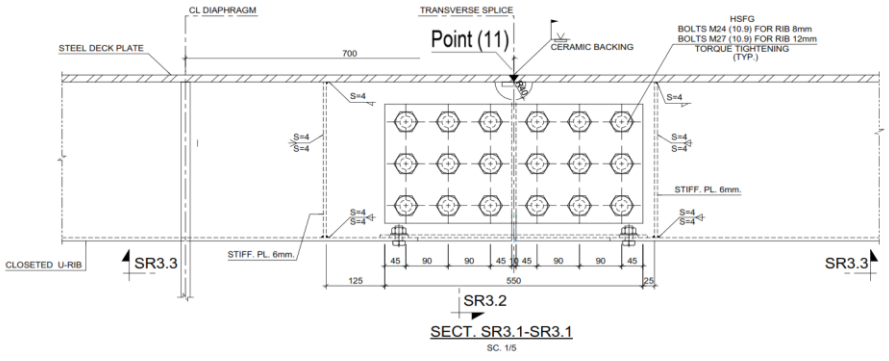
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 0 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 44.7 MPa — 0.00 : U.R

Point (11) — Intermittient Long. Welc — Zones containing cope holes

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.}$ = 79.2 mm

f_{fn} = 52 Mpa Table (19-b) v-(15) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 Mpa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 12.9 Mpa < $\mu_r\, f_f / \gamma_{mft} =$ 39.4 Mpa — 0.33 : U.R



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Plate.G : 0.99 : U.R

Bolts : 0.84 : U.R

Splice Straining actions :

M = -113 KN.m

Q = 153.8 KN

P = -108 KN

T = 6.053 KN.m

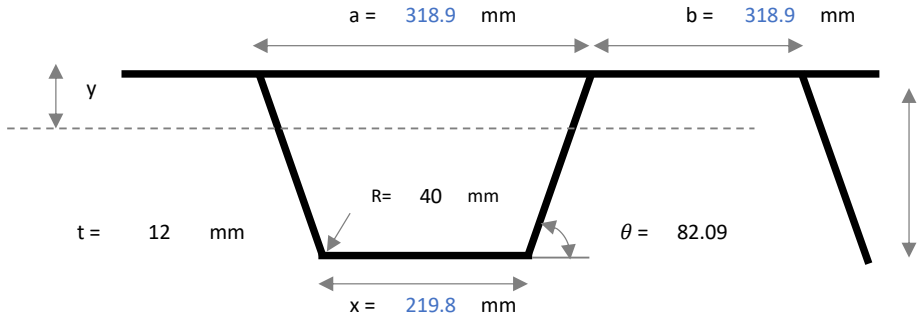
AVERAGE Global Rib stress = -55.26 Mpa

Eq. Global Normal Force = -1160 kN

$N_{Rib\text{-}total}$ = 1268 KN

$F_{y\text{-}actual}$ 350 Mpa

Section Properties :



A = 20997 mm^2

y = 122 mm

I_x = 400631140 mm^4

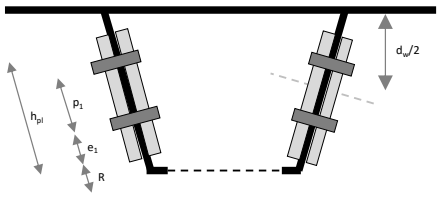
$S_{x\text{-}top}$ = 3293897 mm^3

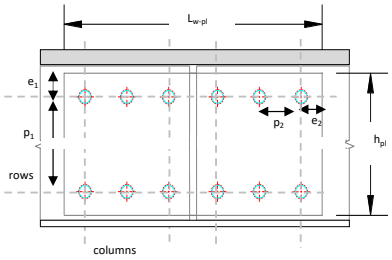
$S_{x\text{-}bot}$ = 1574983 mm^4

$I_{x\text{-}rib}$ = 268650721 mm^4 0.67

Splice plates Dimintions & BOLTES Confiugrations :

Bolts Grade : 10.9





web :

Plates : h_{pl} = 270 mm

< 271 mm (233)

$L_{w\text{-}pl}$ = 532 mm

t_{pl} = 24 mm > 20 mm

> 40 mm

$y_{splice\text{-}web}$ = -13.34 mm $F_{y\ pl.}$ = 330 Mpa

Bolts : M27

No.of Rows 3 $y_{bot\text{-}pl.edge}$ = 43 mm

No.of column 3 $b_{opening}$ 100 mm

e_1 = 41 mm ≥ 1.5 Ø = 40.5 mm

p_1 = 94 mm ≥ 3.0 Ø = 81 mm

e_2 = 40.5 mm ≥ 1.5 Ø = 40.5mm

p_2 = 90 mm ≥ 3.0 Ø = 81 mm

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Check on base material :

tension
compression

(Befor opening net section) :

						$y_{bot-pl.edge}$	y_{Bolt-1}	y_{Bolt-2}	y_{Bolt-3}
A =	18837	mm^2	S_{x-top} =	3313146	mm^3				
y =	113	mm	S_{x-bot} =	1418548	mm^4	43	84	177	270
I_x =	373470022	mm^4							

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ 1268/19 − -113/3.313 = 45 − -23 = 67.7 < $.9f_u/1.25$ 352.8 OK

U.R : 0.192

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ 1268/19 − -113/1.419 = 45 + -53 = 8.4 < $.9f_u/1.25$ 352.8 OK

U.R : 0.024

$0.9\,A_n F_u/\,1.25$ =	6646	KN	<	$A_g F_y/\,1.1$ =	6681	KN
$0.9\,Sx_n F_u/\,1.25$ =	500	KN.m	<	$Sx_g F_y/\,1.1$ =	501	KN.m

(After one bolt column at opening section without bolt holes) :

With 0.6667 Force clause-509.6.3.3

						$b_{opening}$
A_{total} =	19797	mm^2	0.94	S_{x-top} =	3022371	mm^3	
y =	107	mm		S_{x-bot} =	1195513	mm^4	100
Ix =	322103396	mm^4		t_{web} =	12	mm	

clause 609.6.1.2

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ 1268/20 − -113/3.022 = 43 − -25 = 67.7 < $f_y/1.1$ = 318.2 OK

U.R : 0.213

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ 1268/20 − -113/1.196 = 43 + -63 = 20.5 < $f_y/1.1$ = 318.2 OK

U.R : 0.064

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(After one bolt column at opening net section with bolt holes) : With 0.6667 Force

A_{total} = 17637 mm^2 $S_{x\text{-}top}$ = 3031147 mm^3 clause-509.6.3.3

y = 95 mm $S_{x\text{-}bot}$ = 1027924 mm^4

Ix = 288621959 mm^4

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)$ = 1268/18 + -113/3.031 = 48 − -25 = 72.8 < $.9f_u/1.25$ 352.8 OK U.R : 0.206

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)$ = 1268/18 + -113/1.028 = 48 + -73 = 25.6 < $.9f_u/1.25$ 352.8 OK U.R : 0.072

WEB :

Actual Force = 177.3 KN 0.16 : U.R

Yielding	<	V_d = Awg Fy /1.1√3 =	1410.8	KN	clause-506.1.2
Shear Rupture	<	V_{dr} = 0.9 Awn Fu /1.25√3 =	1124.4	KN	
Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	4380.7	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	5126.3	KN	

A_{vn} = 8634.1 mm^2 A_{tn} = 6984 mm^2

A_{vg} = 10434 mm^2 A_{tg} = 10584 mm^2

Actual tension Force A_{rib}/A_{tot} = 627.4 KN 0.23 : U.R

Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	2917.0	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	2776.2	KN	

for Rib : A_{vn} = 6984 mm^2 A_{tn} = 3072 mm^2 t= 12 mm

A_{vg} = 10584 mm^2 A_{tg} = 4512 mm^2

for Splice Plate : A_{vn} = 17568 mm^2 A_{tn} = 12288 mm^2 $2xt_{pl}$ = 48 mm

A_{vg} = 21168 mm^2 A_{tg} = 18048 mm^2

Elliptical BSR :

$\left(\frac{V}{\phi R_{bsv}}\right)^2+\left(\frac{H}{\phi R_{bsh}}\right)^2$ = 0.0405 + 0.226 = 0.26 < 1 0.26 : U.R

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Check on Splice plates :

rib :

Area web splice Agross = 25920 mm^2 ≥ 1.05 Area rib = 10910.30314 mm^2 clause-509.6.3.3 — 0.42 : U.R

Section :

Ix splice = 422800931.1 mm^4 ≥ 1.05 Ix Section = 420662697 mm^2 — 0.99 : U.R

Check on Splice Bolts :

WEB :

$M_{x\text{-web}}$ =	37.991	KN.m	No.of Rows	3	p1 =	94	mm
$M_{due\ to\ shear}$ =	6.9861	KN.m	No.of column	3	p2 =	90	mm
$N_{one\text{-}web}$ =	313.7	KN	no.of Bolts =	9	$X_{C.G}$ =	90	mm
$V_{Torsion} = T/b_{flange}$ =	23.523	KN			$Y_{C.G}$ =	94	mm
$V_{shear} + V_{Torsion}$ =	100.41	KN					
0.5 $M_{x\text{-capacity-rib}}$ =	168.02	KN.m					

$Q_{v\text{-}y}$ =	11.157	KN	$Q_{T\text{-}y}$ =	39.836	KN	Q_y =	50.993	KN
$Q_{N\text{-}x}$ =	34.856	KN	$Q_{T\text{-}x}$ =	41.607	KN	Q_x =	76.462	KN

1.05 Q_{total} =	96.501	KN	one bolt	<	$V_{nsf}/\gamma = \mu$ n K F/γ =	257.04	KN	μ =	0.5
0.5 capacity-rib =	215.18	KN		<					

F_o =	$A_{nb}\ f_o$ =	321.3	KN	n =	2
f_o =	700	N/mm^2		K_h =	1
A_{nb} =	459	mm^2		γ =	1.25

0.84 : U.R

v= 95.58

t= 13.28

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dar making progress together		Date	11/10/2025
		computed by	O.T
	1- Rib 12mm thickness - [-ve σbot. - Hogging] - [P200-Cut]	Approved by	R.AZ

Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490	N/mm^2		
span =	2800	mm		

Straining actions :

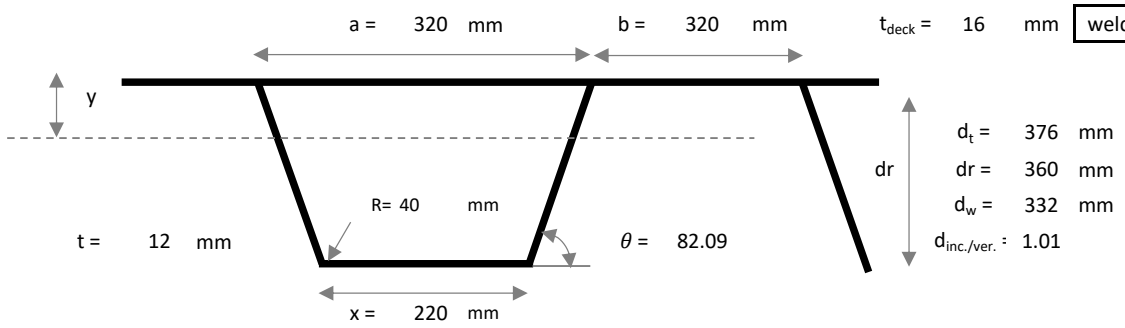
Hogging

M_x =	-137	KN.m
V_y =	195.2	KN
Axial=	-85	KN
T =	5.942	KN.m

Refrences

Ult.	0.29 : U.R
Fatigue	0.50 : U.R
weld	0.83 : U.R

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Hogging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	-4.04	-	-41.5	=	37	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-4.04	+	-87	=	-91	MPa	compression -ve

Local buckling of top Flanges : tension +ve - No check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

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ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-}actual}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

bot. flange : c_3 = 140 mm c_3/t= 11.67 ≤ 42∈ = 34.42

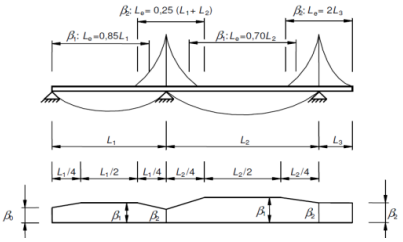
t = 12 mm - No local buckling

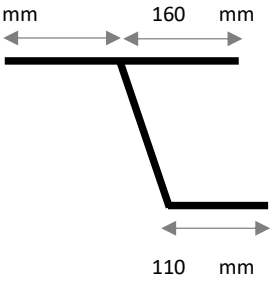
k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\varepsilon\sqrt{k})$= 0.251 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1 ψ = 1 ≥ -3

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Hogging

Le = 0.5 L = 1400 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm k = α bo / Le = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o2} = 160 mm k = α bo / Le = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o3} = 110 mm k = α bo / Le = 0.079 > 0.02

without Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.69 $b_{eff} = B^k b_{c\text{-}eff}$ = 106.8 mm

b_{red} = 3.169 mm

	Design Calculation Sheet	Project no:	IN 18036-0100D
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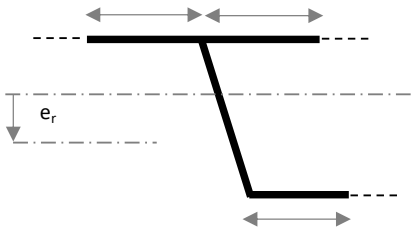
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 9.268 mm b_{red} = 9.268 mm

150.7 mm 150.7 mm



$A_{red\text{-}up}$ = 593.2 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 76.06 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20365.4 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 2.377 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

s_{top} = 3139228 mm^3

s_{bot} = 1541946.7 mm^3

Z= 2319809.0 mm^3 Yu= 16.8 mm

Z_f= 990839.4 mm^3 Yu= 10.13 mm w.o web

b_{red} = 3.169 mm 106.8 mm

t_r = 12 mm

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 335.2 mm t = 12 mm

c/t = 27.93 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

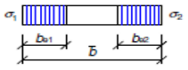
ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\sqrt{k}} =$$ 0.6 ≤ 0.673 ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1.00

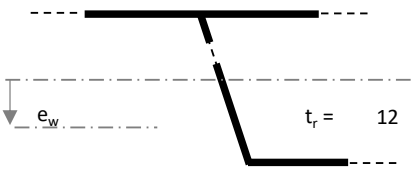
b_c = 335.2 mm b_{eff} = 335.2 mm $y_{red\text{-}top\text{-}web}$= 167.6 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -59.7 mm

A_{new} = 20365.4 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

y_{top} = 123.9 mm s_{top} = 3139228.0 mm^3

$y_{bot.}$ = 252.1 mm s_{bot} = 1541946.7 mm^3

	Design Calculation Sheet	Project no:	IN 18036-0100D
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		computed by	O.T
	1- Rib 12mm thickness - [-ve σbot. - Hogging] - [P200-Cut]	Approved by	R.AZ

Stresses on Effective Section :

σ_{top} = N/A + M/S =	-4.17	-	-43.7	=	39	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-4.17	+	-88.9	=	-93	MPa	compression -ve

Final Effective Section Properties for sagging :

A_{new} =	20365	mm2	y_{top} =	123.9	mm	S_{top} =	3139228 mm^3
I_x =	388799069	mm^3	$y_{bot.}$ =	252.1	mm	S_{bot} =	1541947 mm^3

Check stresses :

IRC 24-2010 clause 510.2

Stress due to mx :

$M_x + N e_y$ = -137 KN.m | $F_{y\text{-actual}}$ 350 MPa | $M_{xdv\text{-top}}$: 998.8 KN.m OK | 0.14 : U.R

$F_{y\text{-actual}}$ 350 MPa | $M_{xdv\text{-bot}}$: 490.6 KN.m OK | 0.28 : U.R

Stress due to Vy :

IRC 24-2010 clause 509.4

d = 363.5 mm | t = 12.0 mm | $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN | $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN | $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1301 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.394 ≤ 0.8 | τ_b = $F_{yw}/(\sqrt{3})$ 202.1 → 202.1 MPa

≤ 1.2 | $[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 267.7

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1301

Check shear Stress

d/t_w = 27.93 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 | no web shear buckling Vn = Vp

OK

V_n = ##### KN | $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 195.22 KN | 0.12 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 22.381 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T/(2A_o t_{deck})$ =	1.988	MPa	<	fd x=	183.7	MPa	0.01 : U.R
$fv_{rib} = T/(2A_o t_{rib})$ =	2.651	MPa	<	fd y=	183.7	MPa	
$fv_{rib\text{-}total}$ =	25.03	MPa	<	fd y=	183.7	MPa	0.14 : U.R

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Axial stress : Axial= -85 KN comp. < Pd= 6318.4 KN OK — 0.01 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.26 < 180 OK $\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}} =$ 0.27

Buckling Class b α = 0.34 $\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2] =$ 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right] =$ 0.98 $f_{crx} = \chi f_y / \gamma_{mo}$ = 310.3 MPa

Ae= 20365.4 mm^2 P_d= 6318.4 KN

-Tension : Ae= 20365.4 mm^2 T_d= 6479.9 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} =$ $\frac{-85}{6318} - \frac{-137}{998.8} =$ -0.01 − -0.14 = 0.12 < 1.0 OK — 0.12 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} =$ $\frac{-85}{6318} + \frac{-137}{490.6} =$ -0.01 + -0.28 = 0.29 < 1.0 OK — 0.29 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} =$ 0.01 + -0.28 = -0.27 < 1.0 OK — clause 510.3.2, -0.27 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$$

Top Deck :	$\frac{39.5^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.0^2}{fy/\gamma_{mo}}$	=	$\frac{39.6}{318.2}$	= 0.12	≤ 1.00	OK — 0.12 : U.R
Top Rib Web :	$\frac{33.9^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{25.0^2}{fy/\gamma_{mo}}$	=	$\frac{55.0}{318.2}$	= 0.17	≤ 1.00	OK — 0.17 : U.R
Bott. Rib Web :	$\frac{80.5^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{25.0^2}{fy/\gamma_{mo}}$	=	$\frac{91.4}{318.2}$	= 0.29	≤ 1.00	OK — 0.29 : U.R
Bott. Rib Flange	$\frac{84.7^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.7^2}{fy/\gamma_{mo}}$	=	$\frac{84.9}{318.2}$	= 0.27	≤ 1.00	OK — 0.27 : U.R

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Fatigue : 0.50 : U.R

Input Data :

γ_{mft} = 1.15

γ_{fft} = 1

N_{sc} = 10 $x10^6$ cycle

Straining actions :

M_x (+)= 53.99 KN.m

Axial= 3.66 KN

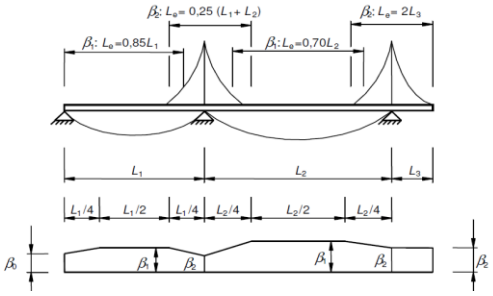
V_y = 84.85 KN

T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

Hogging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :

due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm $b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm

b_{red} = 65.08 mm b_{red} = 65.08 mm

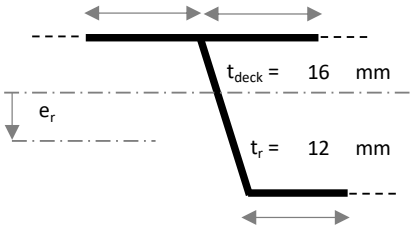
$A_{red\text{-}up}$ = 4165 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 820.3 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 16049.3 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 16.75 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 292368927 mm^4



s_{top} = 2115218 mm^3 $y_{red.cg}$ = 138.2 mm

s_{bot} = 1229586.1 mm^3

$b_{eff} = Bb_{c\text{-}eff}$ = 75.82 mm

b_{red} = 34.18 mm

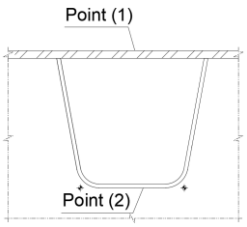
Shear:

A_v = 8723 mm^2

Torsion:

A_o = 93396 mm^2

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Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 138.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.29 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 237.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 44.14 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.49 : U.R

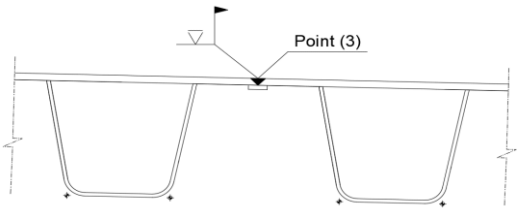
fv x (torsion)= 0.353 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 138.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 72.26 MPa

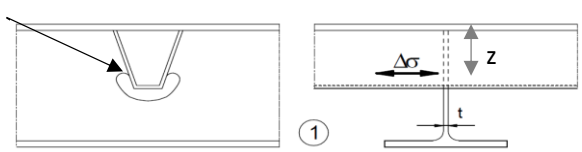
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 62.83 MPa 0.41 : U.R



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Point (4) web at cutout (local effects) Rib web at cutout

z = 230 mm



$\mu_r = \left(\frac{25}{t_f}\right)^{0.25}$ = 1.00 Y = 91.8 mm

f_{fn} = 71 MPa Euro Table (8-8) (1) N=2x10^6

f_{fn} = 52.31 MPa N=5x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 45.54 MPa

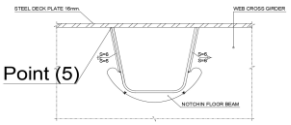
τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 17.18 MPa < $\mu_r f_f / \gamma_{mft}$ = 39.6 MPa 0.43 : U.R

$f_{v\,y}$ = 9.727 MPa $f_{v\,y\,(torsion)}$ = 0.353 MPa

$f_{v\,(total)}$ = 10.08 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Point (5) web weld (longitudinal stresses) Fillet weld one side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25}$ = 1.00 Y = 122.2 mm

f_{fn} = 66 MPa Table (19-b) iii-(13) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 52.38 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 22.8 MPa < $\mu_r f_f / \gamma_{mft}$ = 45.55 MPa 0.50 : U.R

$f_{v\,y}$ = 9.727 MPa $f_{v\,y\,(torsion)}$ = 0.353 MPa

$f_{v\,(total)}$ = 10.08 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Design of weld between Rib and Floor beam :

Q= 205 KN Contact length= 275 mm f_{wn} = $f_u/\sqrt{3}$ γmw= 1.25 $f_{wd}=f_{wn}/\gamma mw$ = 226.3 MPa

$\tau_{shear\ flow} = Q\,S_x/I_x$ = 747.03 N/mm' t_{top} = 12 mm a= $\tau/4f$= 1.43 mm

min S_{weld} = 5 mm s= 2.02 mm use S= 6.0 mm 0.83 : U.R

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	1- Rib 12mm thickness - [-ve σbot. - Sagging] - [P200-Cut]	Approved by	R.AZ

Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490	N/mm^2		
span =	2800	mm		

Straining actions :

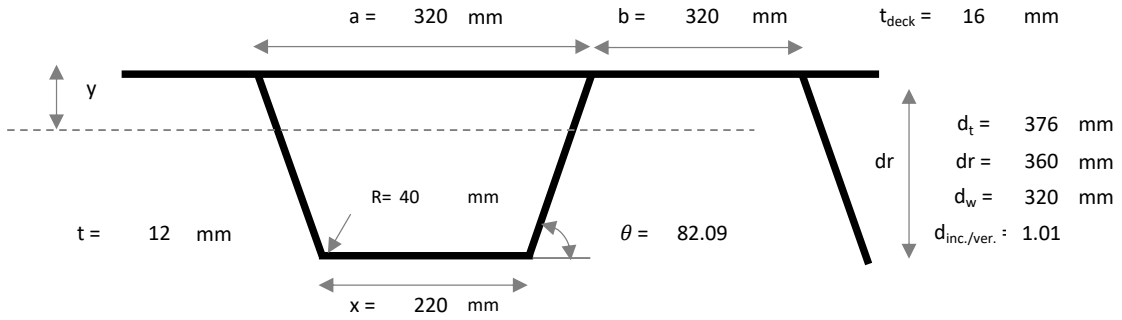
Sagging

M_x =	-113	KN.m
V_y =	153.8	KN
Axial=	-108	KN
T =	6.053	KN.m

Refrences

Ult.	0.24 : U.R
Fatigue	0.56 : U.R
Splice	0.99 Plate
	0.84 Bolts

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Sagging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	-5.12	-	-34.3	=	29	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-5.12	+	-71.9	=	-77	MPa	compression -ve

Local buckling of top Flanges : tension +ve - No check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No check local buckling

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dar making progress together		Date	11/10/2025
		computed by	O.T
	1- Rib 12mm thickness - [-ve σbot. - Sagging] - [P200-Cut]	Approved by	R.AZ

ρ = 1

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-}actual}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

bot. flange : c_3 = 140 mm c_3/t= 11.67 ≤ 42ϵ = 34.42

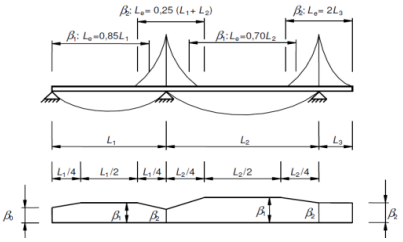
t = 12 mm - No local buckling

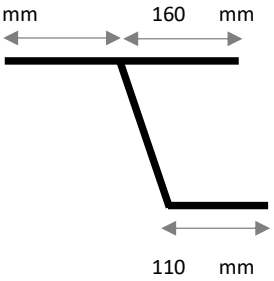
k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\varepsilon\sqrt{k})$= 0.251 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1 ψ = 1 ≥ -3

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Sagging

Le = 0.7 L = 1960 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o2} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o3} = 110 mm k = α bo / Le = 0.056 > 0.02

without Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.98 $b_{eff} = B^k b_{c\text{-}eff}$ = 109.9 mm

b_{red} = 0.123 mm

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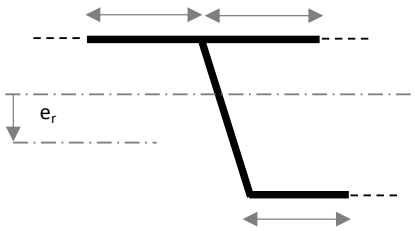
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 0.545 mm b_{red} = 0.545 mm

159.5 mm 159.5 mm



$A_{red\text{-}up}$ = 34.85 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 2.955 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20996.8 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 0.153 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

s_{top} = 3293897 mm^3 y_{top} = 121.6 mm b_{red} = 0.123 mm 109.9 mm

s_{bot} = 1574983.3 mm^3 t_r = 12 mm

Z= 2255736.7 mm^3 Yu= 16.15 mm

Z_f= 1016269.7 mm^3 Yu= 10.07 mm w.o web

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 323.1 mm t = 12 mm

c/t = 26.92 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

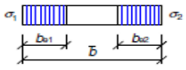
ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\,\sqrt{k}} = 0.578 \le 0.673 \qquad \rho = (\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2 = 1.00$$

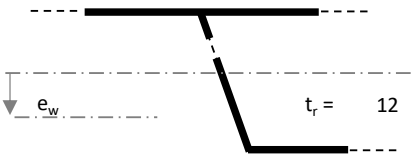
b_c = 323.1 mm b_{eff} = 323.1 mm $y_{red\text{-}top\text{-}web}$ = 161.5 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -55.9 mm

A_{new} = 20996.8 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

y_{top} = 121.6 mm s_{top} = 3293896.7 mm^3

$y_{bot.}$ = 254.4 mm s_{bot} = 1574983.3 mm^3

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Stresses on Effective Section :

σ_{top} = N/A + M/S =	-5.13	-	-34.4	=	29	MPa	tension +ve
$\sigma_{bot.}$ = N/A + M/S =	-5.13	+	-71.9	=	-77	MPa	compression -ve

Final Effective Section Properties for sagging :

A_{new} =	20997	mm^2	y_{top} =	121.6	mm	S_{top} =	3293897	mm^3
I_x =	400631140	mm^4	$y_{bot.}$ =	254.4	mm	S_{bot} =	1574983	mm^3

Check stresses :

IRC 24-2010 clause 510.2

Stress due to mx :

$M_x + N e_y$ =	-113	KN.m	$F_{y\text{-actual}}$	350	MPa	$M_{xdv\text{-top}}$:	1048	KN.m	OK	0.11 : U.R
			$F_{y\text{-actual}}$	350	MPa	$M_{xdv\text{-bot}}$:	501.1	KN.m	OK	0.23 : U.R

Stress due to Vy :

IRC 24-2010 clause 509.4

d = 363.5 mm t = 12.0 mm $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1401 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.38 ≤ 0.8 ≤ 1.2

τ_b = $F_{yw}/(\sqrt{3})$ 202.1 202.1 MPa

$[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 270

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1401

Check shear Stress

d/t_w = 26.92 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 no web shear buckling Vn = Vp

OK

V_n = ##### KN $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 153.77 KN 0.10 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 17.629 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T / (2A_o t_{deck})$ =	2.025	MPa	<	fd x=	183.7	MPa	0.01 : U.R
$fv_{rib} = T / (2A_o t_{rib})$ =	2.701	MPa	<	fd y=	183.7	MPa	
$fv_{rib\text{-total}}$ =	20.33	MPa	<	fd y=	183.7	MPa	0.11 : U.R

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Axial stress : Axial= -108 KN comp. < Pd= 6514.1 KN OK — 0.02 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.27 < 180 OK

$$\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}} = 0.27$$

Buckling Class b α = 0.34

$$\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2] = 0.548$$

$$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right] = 0.98$$

$f_{crx} = \chi f_y / \gamma_{mo}$ = 310.2 MPa

Ae= 20996.8 mm^2 P_d= 6514.1 KN

-Tension : Ae= 20996.8 mm^2 T_d= 6680.8 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{-108}{6514} - \frac{-113}{1048} = -0.02 - 0.11 = 0.09 < 1.0 \text{ OK}$$

0.09 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{-108}{6514} + \frac{-113}{501.1} = -0.02 + -0.23 = 0.24 < 1.0 \text{ OK}$$

0.24 : U.R

-Overall member strength: K_x = 1.00

$$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} = 0.02 + -0.23 = -0.21 < 1.0 \text{ OK}$$

clause 510.3.2 — -0.21 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$$

Top Deck : $\left(\frac{29.3}{fy/\gamma_{mo}}\right)^2 + \left(\frac{0.0}{fy/\gamma_{mo}}\right)^2 - \frac{0.0}{fy/\gamma_{mo}} + 3x\left(\frac{2.0}{fy/\gamma_{mo}}\right)^2 = \frac{29.5}{318.2} = 0.09 \le 1.00$ OK — 0.09 : U.R

Top Rib Web : $\left(\frac{24.7}{fy/\gamma_{mo}}\right)^2 + \left(\frac{0.0}{fy/\gamma_{mo}}\right)^2 - \frac{0.0}{fy/\gamma_{mo}} + 3x\left(\frac{20.3}{fy/\gamma_{mo}}\right)^2 = \frac{43.0}{318.2} = 0.14 \le 1.00$ OK — 0.14 : U.R

Bott. Rib Web : $\left(\frac{63.4}{fy/\gamma_{mo}}\right)^2 + \left(\frac{0.0}{fy/\gamma_{mo}}\right)^2 - \frac{0.0}{fy/\gamma_{mo}} + 3x\left(\frac{20.3}{fy/\gamma_{mo}}\right)^2 = \frac{72.5}{318.2} = 0.23 \le 1.00$ OK — 0.23 : U.R

Bott. Rib Flange $\left(\frac{66.8}{fy/\gamma_{mo}}\right)^2 + \left(\frac{0.0}{fy/\gamma_{mo}}\right)^2 - \frac{0.0}{fy/\gamma_{mo}} + 3x\left(\frac{2.7}{fy/\gamma_{mo}}\right)^2 = \frac{67.0}{318.2} = 0.21 \le 1.00$ OK — 0.21 : U.R

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Fatigue : 0.56 : U.R

Input Data :

γ_{mft} = 1.15
γ_{fft} = 1
N_{sc} = 10 $x10^6$ cycle
Impact Magnification factor : 1.000

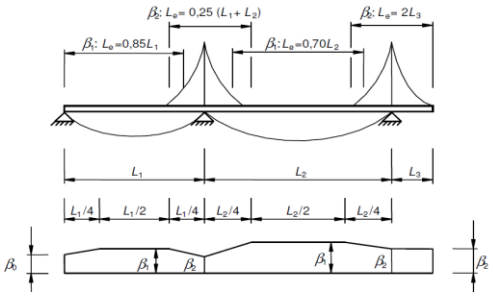
Straining actions :

M_x (+)= 46.57 KN.m
Axial= 2.072 KN
V_y = 81.54 KN
T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

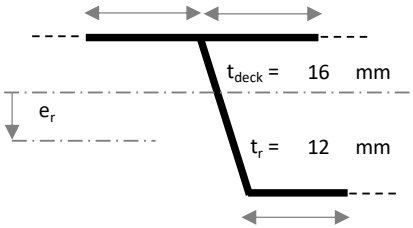
Sagging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :
due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm $b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm
b_{red} = 6.545 mm b_{red} = 6.545 mm

$A_{red\text{-}up}$ = 418.9 mm^2 $y_{red\text{-}olc}$ 113.5 mm
$A_{red\text{-}bot}$ = 52.17 mm^2 $y_{red\text{-}olc}$ 248.5 mm
A_{new} = 20563.6 mm^2
$e_r = (A_{red}\, y_{old})/A_{new}$ = 1.681 mm



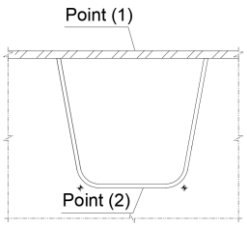
$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 392580543 mm^4

s_{top} = 3187672 mm^3 $y_{red.cg}$ = 123.2 mm $b_{eff} = Bb_{c\text{-}eff}$ = 107.8 mm
s_{bot} = 1552658.3 mm^3 b_{red} = 2.174 mm

Shear:
A_v = 8723 mm^2

Torsion:
A_o = 93396 mm^2

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Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 123.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 14.71 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.16 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 252.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 30.1 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.34 : U.R

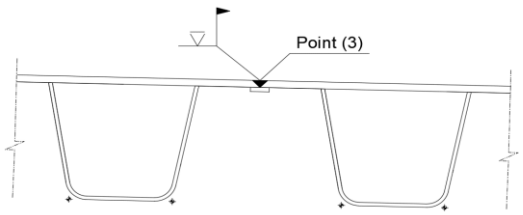
fv x (torsion)= 0.353 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 123.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 72.26 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 14.71 MPa < $\mu_r f_f / \gamma_{mft}$ = 62.83 MPa 0.23 : U.R



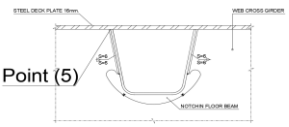
Point (4) web at cutout (local effects) Rib web at cutout

Not applicable

0.00 : U.R

0.00 : U.R

Point (5) web weld (longitudinal stresses) Butt & Fillet weld two side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 107.2 mm

f_{fn} = 92 MPa Table (19-b) i-(9) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 73.02 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 12.81 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 63.5 MPa 0.20 : U.R

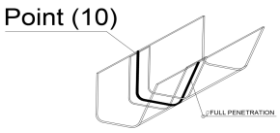
fv y= 9.348 MPa fv y (torsion)= 0.353 MPa

$f_{v\,y(total)}$ = 9.70 MPa < $\mu_r\, \tau_f / \gamma_{mft}$ = 56.7 MPa 0.17 : U.R

Point (10) Welded splice Welded rib splice

Straining actions of Welded splice at location of 0.7m from F.B

Mx =	N
37.4	15
kN.m	kN



$\mu_r = \left(\frac{25}{t_r}\right)^{0.25}$ =1.000 $Y_{bot.}$ = 252.8 mm

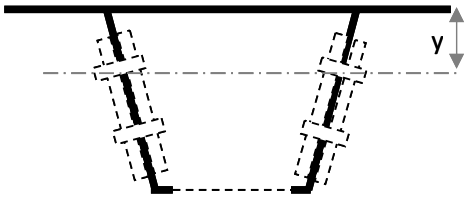
f_{fn} = 80 MPa Euro Table (8-8) (4) N=2x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 51.31 MPa

f_{fn} = 58.94 MPa N=5x10^6

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 24.82 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 44.62 MPa 0.56 : U.R

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	1- Rib 12mm thickness - [-ve σbot. - Sagging] - [P200-Cut]	Approved by	R.AZ

Section 2-2 (Splice): Straining action of fatigue at Splice



A =	17637	mm^2	$S_{x\text{-top}}$ =	3031147	mm^3
y =	95	mm	$S_{x\text{-bot}}$ =	1027924	mm^4
I_x =	288621959	mm^4			

web

No.of rows	3
Hole Diameter	30

Point (6) Upper deck Plate Butt weld one side (Transverse)

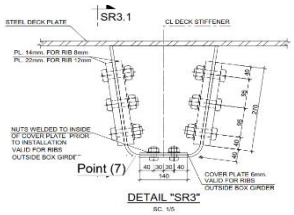
$$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} = 1.000 \qquad Y_{yp.} = 95.2 \text{ mm}$$

f_{fn} = 52 MPa Table (19-b) ix-(23) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} = 45.3$ MPa

$$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 15.5 \text{ MPa} \quad < \quad \mu_r f_f / \gamma_{mft} = 39.4 \text{ MPa}$$

0.39 : U.R

Point (7) Bottom Flange Plate Net Section (Bollted)

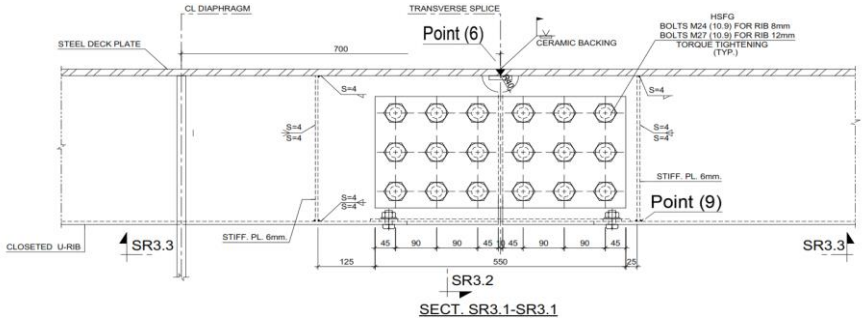


$$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000 \qquad Y_{bot.} = 281 \text{ mm}$$

f_{fn} = 103 MPa Table (19-a) ii-(5) $f_f = f_{fn} \sqrt[5]{\frac{5x\,10^6}{N_{sc}}} = 89.7$ MPa

$$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 45.4 \text{ MPa} \quad < \quad \mu_r f_f / \gamma_{mft} = 78 \text{ MPa}$$

0.58 : U.R



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Point (9) for sagging — Web Vertical stiffeners — Vertical stiffeners Fillet welded

Straining actions and gross section

Mx =	N
0	0

Bolted splice at location of 0.7m and closure plate at 1.0m from F.B

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.} = 240.8$ mm

$f_{fn} =$ 59 MPa Table (19-b) xviii-(36&37) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 51.4 MPa

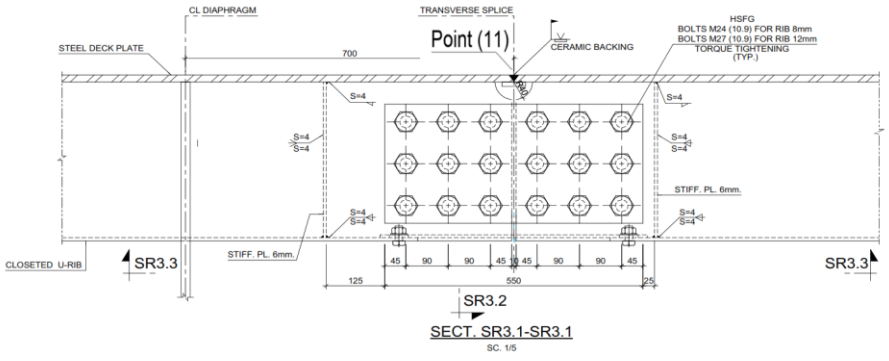
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 0 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 44.7 MPa — 0.00 : U.R

Point (11) — Intermittient Long. Welc — Zones containing cope holes

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.} = 79.2$ mm

$f_{fn} =$ 52 Mpa Table (19-b) v-(15) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 Mpa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 12.9 Mpa < $\mu_r\, f_f / \gamma_{mft} =$ 39.4 Mpa — 0.33 : U.R



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Plate.G : 0.99 : U.R

Bolts : 0.84 : U.R

Splice Straining actions :

M = -113 KN.m

Q = 153.8 KN

P = -108 KN

T = 6.053 KN.m

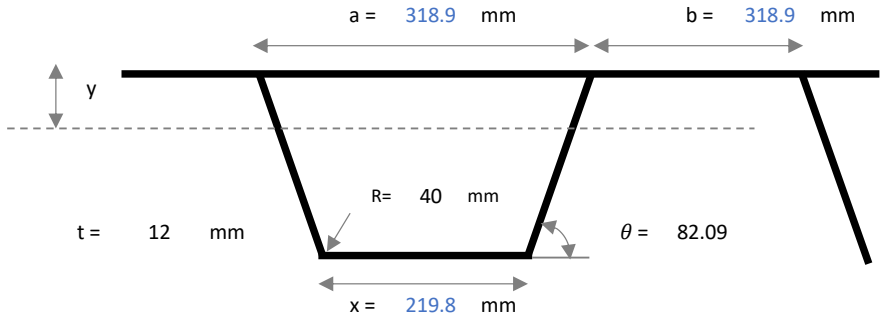
AVERAGE Global Rib stress = -55.26 Mpa

Eq. Global Normal Force = -1160 kN

$N_{Rib\text{-}total}$ = 1268 KN

$F_{y\text{-}actual}$ 350 Mpa

Section Properties :



A = 20997 mm^2

y = 122 mm

I_x = 400631140 mm^4

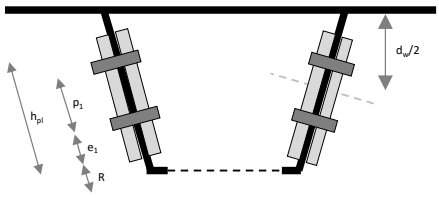
$S_{x\text{-}top}$ = 3293897 mm^3

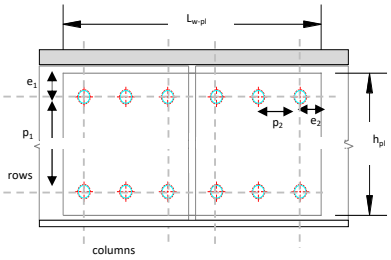
$S_{x\text{-}bot}$ = 1574983 mm^4

$I_{x\text{-}rib}$ = 268650721 mm^4 0.67

Splice plates Dimintions & BOLTES Confiugrations :

Bolts Grade : 10.9





web :

Plates : h_{pl} = 270 mm

< 271 mm

$L_{w\text{-}pl}$ = 532 mm

t_{pl} = 24 mm > 20 mm

> 40 mm

$y_{splice\text{-}web}$ = -13.34 mm $F_{y\ pl.}$ = 330 Mpa

Bolts : M27

No.of Rows 3 $y_{bot\text{-}pl.edge}$ = 43 mm

No.of column 3 $b_{opening}$ 100 mm

e_1 = 41 mm ≥ 1.5 Ø = 40.5 mm

p_1 = 94 mm ≥ 3.0 Ø = 81 mm

e_2 = 40.5 mm ≥ 1.5 Ø = 40.5mm

p_2 = 90 mm ≥ 3.0 Ø = 81 mm

dar making progress together	Design Calculation Sheet	Project no:	IN 18036-0100D
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Check on base material :

tension
compression

(Befor opening net section) :

						$y_{bot-pl.edge}$	y_{Bolt-1}	y_{Bolt-2}	y_{Bolt-3}
A =	18837	mm^2	S_{x-top} =	3313146	mm^3				
y =	113	mm	S_{x-bot} =	1418548	mm^4	43	84	177	270
I_x =	373470022	mm^4							

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1268}{19}-\frac{-113}{3.313}$ = 45 − -23 = 67.7 < $.9f_u/1.25$ 352.8 OK

U.R : 0.192

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1268}{19}-\frac{-113}{1.419}$ = 45 + -53 = 8.4 < $.9f_u/1.25$ 352.8 OK

U.R : 0.024

$0.9\,A_n F_u/\,1.25$ =	6646	KN	<	$A_g F_y/\,1.1$ =	6681 KN
$0.9\,Sx_n F_u/\,1.25$ =	500	KN.m	<	$Sx_g F_y/\,1.1$ =	501 KN.m

(After one bolt column at opening section without bolt holes) :

With 0.6667 Force clause-509.6.3.3

						$b_{opening}$
A_{total} =	19797	mm^2	0.94	S_{x-top} =	3022371	mm^3	
y =	107	mm		S_{x-bot} =	1195513	mm^4	100
Ix =	322103396	mm^4		t_{web} =	12	mm	

clause 609.6.1.2

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1268}{20}-\frac{-113}{3.022}$ = 43 − -25 = 67.7 < $f_y/1.1$ = 318.2 OK

U.R : 0.213

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1268}{20}-\frac{-113}{1.196}$ = 43 + -63 = 20.5 < $f_y/1.1$ = 318.2 OK

U.R : 0.064

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(After one bolt column at opening net section with bolt holes) : With 0.6667 Force

A_{total} = 17637 mm^2 $S_{x\text{-}top}$ = 3031147 mm^3 clause-509.6.3.3

y = 95 mm $S_{x\text{-}bot}$ = 1027924 mm^4

Ix = 288621959 mm^4

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)$ = 1268/18 − -113/3.031 = 48 − -25 = 72.8 < .9f_u/1.25 352.8 OK U.R : 0.206

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)$ = 1268/18 − -113/1.028 = 48 + -73 = 25.6 < .9f_u/1.25 352.8 OK U.R : 0.072

WEB :

Actual Force = 177.3 KN 0.16 : U.R

Yielding	<	V_d = Awg Fy /1.1√3 =	1410.8	KN	clause-506.1.2
Shear Rupture	<	V_{dr} = 0.9 Awn Fu /1.25√3 =	1124.4	KN	
Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	4380.7	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	5126.3	KN	

A_{vn} = 8634.1 mm^2 A_{tn} = 6984 mm^2

A_{vg} = 10434 mm^2 A_{tg} = 10584 mm^2

Actual tension Force A_{rib}/A_{tot} = 627.4 KN 0.23 : U.R

Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	2917.0	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	2776.2	KN	

for Rib : A_{vn} = 6984 mm^2 A_{tn} = 3072 mm^2 t= 12 mm

A_{vg} = 10584 mm^2 A_{tg} = 4512 mm^2

for Splice Plate : A_{vn} = 17568 mm^2 A_{tn} = 12288 mm^2 $2xt_{pl}$ = 48 mm

A_{vg} = 21168 mm^2 A_{tg} = 18048 mm^2

Elliptical BSR :

$\left(\frac{V}{\phi R_{bsv}}\right)^2+\left(\frac{H}{\phi R_{bsh}}\right)^2$ = 0.0405 + 0.226 = 0.26 < 1 0.26 : U.R

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Check on Splice plates :

rib :

Area web splice Agross = 25920 mm^2 ≥ 1.05 Area rib = 10910.30314 mm^2 clause-509.6.3.3 — 0.42 : U.R

Section :

Ix splice = 422800931.1 mm^4 ≥ 1.05 Ix Section = 420662697 mm^2 — 0.99 : U.R

Check on Splice Bolts :

WEB :

$M_{x\text{-web}}$ =	37.991	KN.m	No.of Rows	3	p1 =	94	mm
$M_{due\ to\ shear}$ =	6.9861	KN.m	No.of column	3	p2 =	90	mm
$N_{one\text{-}web}$ =	313.7	KN	no.of Bolts =	9	$X_{C.G}$ =	90	mm
$V_{Torsion} = T/b_{flange}$ =	23.523	KN			$Y_{C.G}$ =	94	mm
$V_{shear} + V_{Torsion}$ =	100.41	KN					
0.5 $M_{x\text{-capacity-rib}}$ =	168.02	KN.m					

$Q_{v\text{-}y}$ =	11.157	KN	$Q_{T\text{-}y}$ =	39.836	KN	Q_y =	50.993	KN
$Q_{N\text{-}x}$ =	34.856	KN	$Q_{T\text{-}x}$ =	41.607	KN	Q_x =	76.462	KN

1.05 Q_{total} =	96.501	KN	one bolt	<	$V_{nsf}/\gamma = \mu$ n K F/γ =	257.04	KN	μ =	0.5
0.5 capacity-rib =	215.18	KN		<					
					F_o = A_{nb} f_o =	321.3	KN	n =	2
					f_o =	700	N/mm^2	K_h =	1
					A_{nb} =	459	mm^2	γ =	1.25

v= 95.58
t= 13.28

0.84 : U.R

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	1- Rib 12mm thickness - [-ve σtop - Hogging] - [P200-Cut]	Approved by	R.AZ

Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490			N/mm^2
span =	2800			mm

Straining actions :

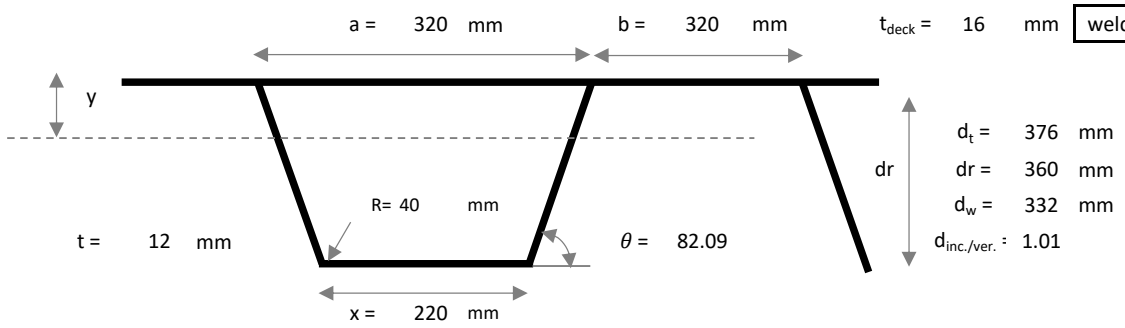
Hogging

M_x =	61.73	KN.m
V_y =	153.6	KN
Axial=	-15.6	KN
T =	5.612	KN.m

Refrences

Ult.	0.16 : U.R
Fatigue	0.50 : U.R
weld	0.83 : U.R

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Hogging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	-0.74	-	18.69	=	-19	MPa	compression -ve
$\sigma_{bot.}$ = N/A + M/S =	-0.74	+	39.16	=	38	MPa	tension +ve

Local buckling of top Flanges : compression -ν - check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\epsilon\sqrt{k})$ = 0.43 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

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k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\epsilon\sqrt{k}) =$ 0.43 ≤ 0.673

ρ = $(\lambda - 0.055(3+\psi))/\lambda^2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

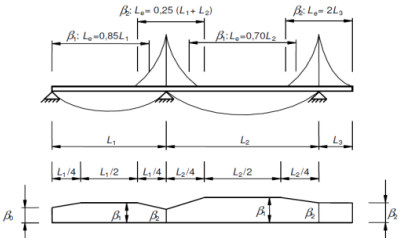
bot. flange : c_3 = 140 mm c_3/t = 11.67 ≤ 42ϵ = 34.42

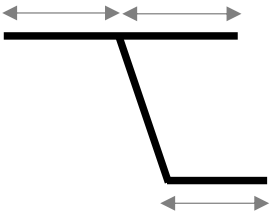
t = 12 mm - No check local buckling

ρ = 1

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Hogging

Le = 0.5 L = 1400 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm $k = \alpha\, bo / Le$ = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o2} = 160 mm $k = \alpha\, bo / Le$ = 0.114 > 0.02

with Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.59 $b_{eff} = B^k b_{c\text{-}eff}$ = 150.7 mm

b_{red} = 9.268 mm

b_{o3} = 110 mm $k = \alpha\, bo / Le$ = 0.079 > 0.02

without Local buckling ≤ 0.7

β2 = 1/(1+6(k-1/2500k)+1.6k2)= 0.69 $b_{eff} = B^k b_{c\text{-}eff}$ = 106.8 mm

b_{red} = 3.169 mm

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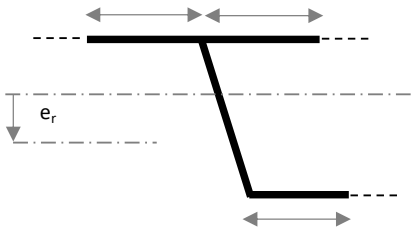
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 9.268 mm b_{red} = 9.268 mm

150.7 mm 150.7 mm



$A_{red\text{-}up}$ = 593.2 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 76.06 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20365.4 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 2.377 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4

s_{top} = 3139228 mm^3 b_{red} = 3.169 mm 106.8 mm

s_{bot} = 1541946.7 mm^3 t_r = 12 mm

Z= 2319809.0 mm^3 Yu= 16.8 mm

Z_f= 990839.4 mm^3 Yu= 10.13 mm w.o web

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 335.2 mm t = 12 mm

c/t = 27.93 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

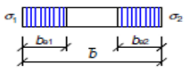
ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\,\sqrt{k}} =$$ 0.6 ≤ 0.673 ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda^2$= 1.00

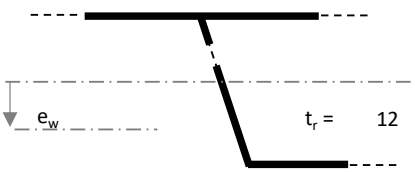
b_c = 335.2 mm b_{eff} = 335.2 mm $y_{red\text{-}top\text{-}web}$= 167.6 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -59.7 mm

A_{new} = 20365.4 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

y_{top} = 123.9 mm s_{top} = 3139228.0 mm^3

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 388799069 mm^4 $y_{bot.}$ = 252.1 mm s_{bot} = 1541946.7 mm^3

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Stresses on Effective Section :

σ_{top} = N/A + M/S =	-0.77	-	19.66	=	-20	MPa	compression -ve
$\sigma_{bot.}$ = N/A + M/S =	-0.77	+	40.03	=	39	MPa	tension +ve

Final Effective Section Properties for sagging :

A_{new} =	20365	mm2	y_{top} =	123.9	mm	S_{top} =	3139228 mm^3
I_x =	388799069	mm^3	$y_{bot.}$ =	252.1	mm	S_{bot} =	1541947 mm^3

Check stresses :

IRC 24-2010 clause 510.2

Stress due to mx :

$M_x + N e_y$ = 61.73 KN.m | $F_{y\text{-actual}}$ 350 MPa | $M_{xdv\text{-top}}$: 998.8 KN.m OK | 0.06 : U.R

$F_{y\text{-actual}}$ 350 MPa | $M_{xdv\text{-bot}}$: 490.6 KN.m OK | 0.13 : U.R

Stress due to Vy :

IRC 24-2010 clause 509.4

d = 363.5 mm | t = 12.0 mm | $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN | $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN | $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1301 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.394 ≤ 0.8 | τ_b = $F_{yw}/(\sqrt{3})$ 202.1 202.1 MPa

≤ 1.2 | $[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 267.7

$F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1301

Check shear Stress

d/t_w = 27.93 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 | no web shear buckling Vn = Vp

OK

V_n = ##### KN | $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 153.6 KN | 0.10 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 17.608 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T/(2A_o t_{deck})$ =	1.878	MPa	<	fd x=	183.7	MPa	0.01 : U.R
$fv_{rib} = T/(2A_o t_{rib})$ =	2.504	MPa	<	fd y=	183.7	MPa	
$fv_{rib\text{-}total}$ =	20.11	MPa	<	fd y=	183.7	MPa	0.11 : U.R

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		computed by	O.T
	1- Rib 12mm thickness - [-ve σtop - Hogging] - [P200-Cut]	Approved by	R.AZ

Axial stress : Axial= -15.6 KN comp. < Pd= 6318.4 KN OK — 0.00 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.26 < 180 OK $\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}}$ = 0.27

Buckling Class b α = 0.34 $\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2]$ = 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right]$ = 0.98 $f_{crx} = \chi f_y / \gamma_{mo}$ = 310.3 MPa

Ae= 20365.4 mm^2 P_d= 6318.4 KN

-Tension : Ae= 20365.4 mm^2 T_d= 6479.9 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{-15.6}{6318} - \frac{61.73}{998.8}$ = 0.00 − 0.06 = 0.06 < 1.0 OK — 0.06 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{-15.6}{6318} + \frac{61.73}{490.6}$ = 0.00 + 0.13 = 0.12 < 1.0 OK — 0.12 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}}$ = 0.00 + 0.06 = 0.06 < 1.0 OK — clause 510.3.2, 0.06 : U.R

-Von misses stresses: — EN-1993-1-1 6.2.1(5)

$\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$

Top Deck :	$\frac{-20.4^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{1.9^2}{fy/\gamma_{mo}}$	= $\frac{20.7}{318.2}$	= 0.07	≤ 1.00	OK	0.07 : U.R
Top Rib Web :	$\frac{-17.9^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{20.1^2}{fy/\gamma_{mo}}$	= $\frac{39.2}{318.2}$	= 0.12	≤ 1.00	OK	0.12 : U.R
Bott. Rib Web :	$\frac{-38.9^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{20.1^2}{fy/\gamma_{mo}}$	= $\frac{52.2}{318.2}$	= 0.16	≤ 1.00	OK	0.16 : U.R
Bott. Rib Flange	$\frac{-40.8^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.5^2}{fy/\gamma_{mo}}$	= $\frac{41.0}{318.2}$	= 0.13	≤ 1.00	OK	0.13 : U.R

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Fatigue : 0.50 : U.R

Input Data :

γ_{mft} = 1.15

γ_{fft} = 1

N_{sc} = 10 $x10^6$ cycle

Straining actions :

M_x (+)= 53.99 KN.m

Axial= 3.66 KN

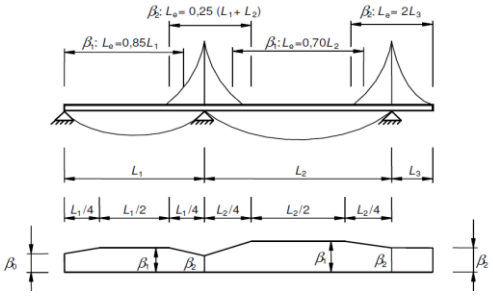
V_y = 84.85 KN

T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

Hogging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :

due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm $b_{eff} = Bb_{c\text{-}eff}$ = 94.92 mm

b_{red} = 65.08 mm b_{red} = 65.08 mm

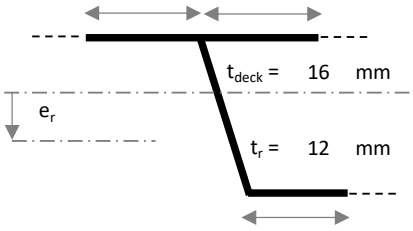
$A_{red\text{-}up}$ = 4165 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 820.3 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 16049.3 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 16.75 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 292368927 mm^4



s_{top} = 2115218 mm^3 $y_{red.cg}$ = 138.2 mm

s_{bot} = 1229586.1 mm^3

$b_{eff} = Bb_{c\text{-}eff}$ = 75.82 mm

b_{red} = 34.18 mm

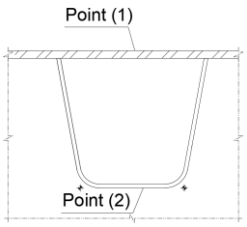
Shear:

A_v = 8723 mm^2

Torsion:

A_o = 93396 mm^2

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Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 138.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.29 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 237.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 44.14 MPa < $\mu_r f_f / \gamma_{mft}$ = 89.33 MPa 0.49 : U.R

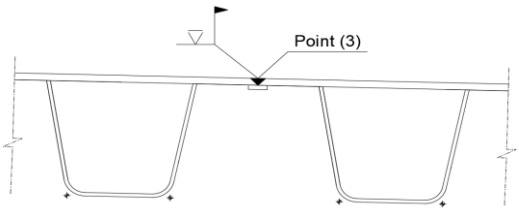
fv x (torsion)= 0.353 MPa < $\mu_r \tau_f / \gamma_{mft}$ = 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 138.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 72.26 MPa

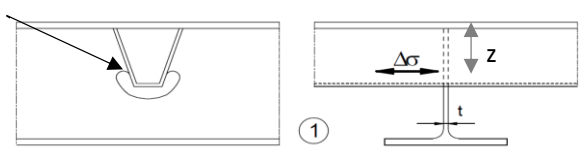
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 25.75 MPa < $\mu_r f_f / \gamma_{mft}$ = 62.83 MPa 0.41 : U.R



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Point (4) web at cutout (local effects) Rib web at cutout

z = 230 mm



$\mu_r = \left(\frac{25}{t_f}\right)^{0.25}$ = 1.00 Y = 91.8 mm

f_{fn} = 71 MPa Euro Table (8-8) (1) N=2x10^6

f_{fn} = 52.31 MPa N=5x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 45.54 MPa

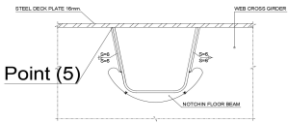
τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 17.18 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 39.6 MPa 0.43 : U.R

fv y= 9.727 MPa fv y (torsion)= 0.353 MPa

$f_{v\,y(total)}$ = 10.08 MPa < $\mu_r\, \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Point (5) web weld (longitudinal stresses) Fillet weld one side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25}$ = 1.00 Y = 122.2 mm

f_{fn} = 66 MPa Table (19-b) iii-(13) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 52.38 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 22.8 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 45.55 MPa 0.50 : U.R

fv y= 9.727 MPa fv y (torsion)= 0.353 MPa

$f_{v\,y(total)}$ = 10.08 MPa < $\mu_r\, \tau_f / \gamma_{mft}$ = 56.7 MPa 0.18 : U.R

Design of weld between Rib and Floor beam :

Q= 205 KN Contact length= 275 mm $f_{wn} = f_u/\sqrt{3}$ γmw= 1.25 $f_{wd} = f_{wn}/\gamma mw$ = 226.3 MPa

$\tau_{shear\ flow} = Q\,S_x/I_x$ = 747.03 N/mm' t_{top} = 12 mm a= $\tau/4f$= 1.43 mm

min S_{weld} = 5 mm s= 2.02 mm use S= 6.0 mm 0.83 : U.R

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Input Date :

	t ≤20	t ≤40	t >40	
F_y =	350	330	320	MPa
Fu =	490	N/mm^2		
span =	2800	mm		

Straining actions :

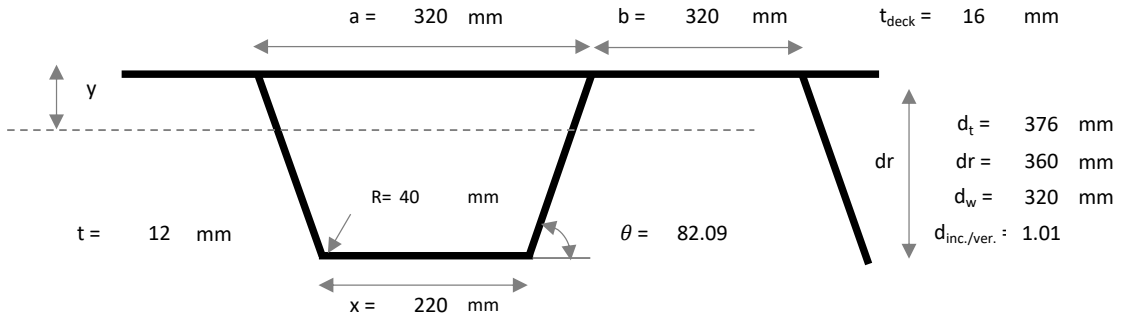
Sagging

M_x =	77.83	KN.m
V_y =	127.4	KN
Axial=	42.95	KN
T =	5.934	KN.m

Refrences

Ult.	0.17 : U.R
Fatigue	0.56 : U.R
Splice	0.99 Plate
	0.84 Bolts

Section Properties :



A =	21034.6	mm^2	I_x =	401263727	mm^4
y_{top} =	121.5	mm	s_{top} =	3303263.1	mm^3
$y_{bot.}$ =	254.5	mm	s_{bot} =	1576519.6	mm^3

Effective Section Properties : Sagging

Stresses on Gross Section :

σ_{top} = N/A + M/S =	2.042	-	23.56	=	-22	MPa	compression -ve
$\sigma_{bot.}$ = N/A + M/S =	2.042	+	49.37	=	51	MPa	tension +ve

Local buckling of top Flanges : compression -ν - check local buckling

$t_{b.F}$ = 16 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819 Euro

top flange : c_1 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\epsilon\sqrt{k})$ = 0.43 ≤ 0.673

ρ = $(\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

c_2 = 320 mm c_1/t = 20 ≤ 42ϵ = 34.42

t = 16 mm - No local buckling

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k = 4 $\lambda_p = \sqrt{(f_y/\sigma)} = (c/t)/(28.4\epsilon\sqrt{k}) =$ 0.43 ≤ 0.673

ρ = $(\lambda - 0.055(3+\psi))/\lambda^2$ = 1 ψ = 1 ≥ -3

b_{eff} = pb = 320 mm b_{red} = 0 mm

Local buckling of bot. Flanges : $t_{b.F}$ = 12 mm ≤ 20 mm $F_{y\text{-actual}}$ 350 MPa

≤ 40 mm $\epsilon_f = \sqrt{(}$ 235 /fy) = 0.819

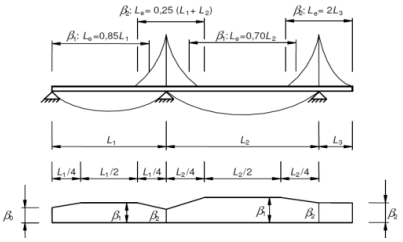
bot. flange : c_3 = 140 mm c_3/t = 11.67 ≤ 42ϵ = 34.42

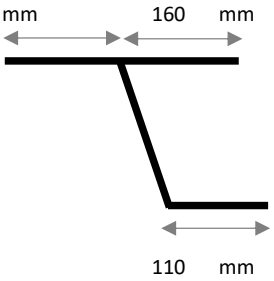
t = 12 mm - No check local buckling

ρ = 1

b_{eff} = pb = 140 mm b_{red} = 0 mm

Shear lag Flanges :





Sagging

Le = 0.7 L = 1960 mm α = 1 no longitudinal stiffeners

b_{o1} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o2} = 160 mm k = α bo / Le = 0.082 > 0.02

with Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.96 $b_{eff} = B^k b_{c\text{-}eff}$ = 159.5 mm

b_{red} = 0.545 mm

b_{o3} = 110 mm k = α bo / Le = 0.056 > 0.02

without Local buckling ≤ 0.7

β1 = 1/(1+6.4k2)= 0.98 $b_{eff} = B^k b_{c\text{-}eff}$ = 109.9 mm

b_{red} = 0.123 mm

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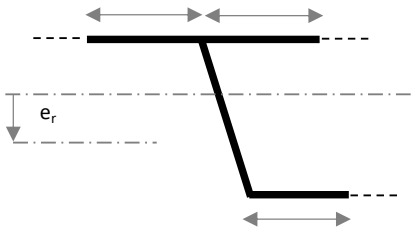
Effective Section Properties for sagging :

due to reduction of flanges

t_{deck} = 16 mm

b_{red} = 0.545 mm b_{red} = 0.545 mm

159.5 mm 159.5 mm



$A_{red\text{-}up}$ = 34.85 mm^2 $y_{red\text{-}olc}$ 113.5 mm

$A_{red\text{-}bot}$ = 2.955 mm^2 $y_{red\text{-}olc}$ 248.5 mm

A_{new} = 20996.8 mm^2

$e_r = (A_{red}\, y_{old})/A_{new}$ = 0.153 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

s_{top} = 3293897 mm^3 y_{top} = 121.6 mm b_{red} = 0.123 mm 109.9 mm

s_{bot} = 1574983.3 mm^3 t_r = 12 mm

Z= 2255736.7 mm^3 Yu= 16.15 mm

Z_f= 1016269.7 mm^3 Yu= 10.07 mm w.o web

Local buckling of Webs (Euro) : $\epsilon = \sqrt{(235/fy)}$ = 0.82

- Considering uniform compression stresses from Global Analysis

ψ = F1 / F2 = 1 > -1 c = 323.1 mm t = 12 mm

c/t = 26.92 ≤ $=42\epsilon/(0.67+0.33\psi)$= 34.4 ≥ -3 - local buckling

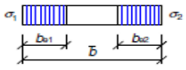
ψ = 1 ≤ 1 k = $8.2 / (1.05 + \psi)$= 4

> 0

$$\lambda_p = \sqrt{\frac{fy}{\sigma}} = \frac{c/t}{28.4\,\epsilon\,\sqrt{k}} = 0.578 \le 0.673 \qquad \rho = (\lambda\text{-}0.055(3+\psi))/\lambda\text{^}2 = 1.00$$

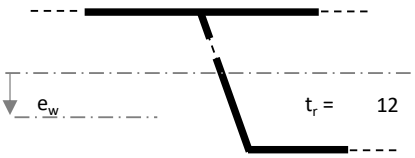
b_c = 323.1 mm b_{eff} = 323.1 mm $y_{red\text{-}top\text{-}web}$= 161.5 mm

b_t = 0 mm b_{red} = 0 mm



Effective Section Properties for sagging :

due to reduction of webs



A_{red} = 0 mm^2 $y_{red\text{-}old.c}$ -55.9 mm

A_{new} = 20996.8 mm^2

$e_w = (A_{red}\, y_{old})/A_{new}$ = 0 mm

$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 400631140 mm^4

y_{top} = 121.6 mm s_{top} = 3293896.7 mm^3

$y_{bot.}$ = 254.4 mm s_{bot} = 1574983.3 mm^3

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Stresses on Effective Section :

σ_{top} = N/A + M/S =	2.046	-	23.63	=	-22	MPa	compression -ve
$\sigma_{bot.}$ = N/A + M/S =	2.046	+	49.41	=	51	MPa	tension +ve

Final Effective Section Properties for sagging :

A_{new} =	20997	mm^2	y_{top} =	121.6	mm	S_{top} = 3293897 mm^3
I_x =	400631140	mm^4	$y_{bot.}$ =	254.4	mm	S_{bot} = 1574983 mm^3

Check stresses :

IRC 24-2010

Stress due to mx :

clause 510.2

$M_x + N\,e_y$ = 77.83 KN.m | $F_{y\text{-actual}}$ 350 MPa | $M_{xdv\text{-top}}$: 1048 KN.m OK | 0.07 : U.R

$F_{y\text{-actual}}$ 350 MPa | $M_{xdv\text{-bot}}$: 501.1 KN.m OK | 0.16 : U.R

Stress due to Vy :

IRC 24-2010

clause 509.4

d = 363.5 mm | t = 12.0 mm | $A_v = 2d_w t_w$ = 8722.9 mm^2

i) Plastic shear resistance $V_p = A_v F_{yw}/\sqrt{3}$ = 1762.7 KN | $F_{y\text{-actual}}$ 350 MPa

ii) Resistance to shear buckling $V_{cr} = A_v \tau_b$ = 1763 KN | $\tau_{cr} = \frac{K_v \pi^2 E}{12(1-\mu^2)(d/tw)^2}$ = 1401 MPa

$\lambda_w = \sqrt{f_y/\tau_{cr}\sqrt{3}}$ = 0.38 ≤ 0.8 ≤ 1.2

τ_b = $F_{yw}/(\sqrt{3})$ 202.1 | $[1-0.8(\lambda w-0.8)](Fy/\sqrt{3})$ = 270 | $F_{yw}/(\sqrt{3}\,\lambda_w^2)$ = 1401 | 202.1 MPa

Check shear Stress

d/t_w = 26.92 ≤ 67 $\varepsilon\sqrt{(K_v/5.35)}$ = 54.9 no web shear buckling Vn = Vp

OK

V_n = ##### KN | $V_d = V_n / \gamma_{mo}$ = 1602 KN > V_y = 127.42 KN | 0.08 : U.R

$f_{d\,y}$ = 183.7 MPa > $f_{v\,y}$ = 14.607 MPa

Stress due to T :

A_o = 93396 mm^2

$fv_{deck} = T/(2A_o t_{deck})$ =	1.985	MPa	<	fd x=	183.7	MPa	0.01 : U.R
$fv_{rib} = T/(2A_o t_{rib})$ =	2.647	MPa	<	fd y=	183.7	MPa	
$fv_{rib\text{-total}}$ =	17.25	MPa	<	fd y=	183.7	MPa	0.09 : U.R

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Axial stress : Axial= 42.95 KN Tension < Pd= 6680.8 KN OK — 0.01 : U.R

-Compression : $F_{y\text{-actual}}$ 350 MPa — clause 506.1

$(KL/r)_x$= 20.27 < 180 OK $\lambda = \sqrt{\frac{F_y}{F_{cr}}} = \left(\frac{KL}{r}\right)\sqrt{\frac{F_y}{\pi^2 E}}$ = 0.27

Buckling Class b α = 0.34 $\emptyset = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2]$ = 0.548

$\chi = \left[\frac{1}{\emptyset + (\emptyset^2 - \lambda^2)^{0.5}}\right]$ = 0.98 $f_{crx} = \chi f_y / \gamma_{mo}$ = 310.2 MPa

Ae= 20996.8 mm^2 P_d= 6514.1 KN

-Tension : Ae= 20996.8 mm^2 T_d= 6680.8 KN — clause 507.1

Combined forces:

-Section strength: — clause 510.3.1

-Top stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{42.95}{6681} - \frac{77.83}{1048} = 0.01 - 0.07 = 0.07 < 1.0$ OK — 0.07 : U.R

-Bottom stresses: $F_{y\text{-actual}}$ = 350 MPa

$\frac{N}{N_d} + \frac{M_x}{M_{dx}} = \frac{42.95}{6681} + \frac{77.83}{501.1} = 0.01 + 0.16 = 0.16 < 1.0$ OK — 0.16 : U.R

-Overall member strength: K_x = 1.00

$\frac{N}{N_{dx}} + kx\frac{C_{mx}M_x}{M_{dx}} =$ 0.00 + 0.07 = 0.07 < 1.0 OK — clause 510.3.2, 0.07 : U.R

-Von misses stresses: $\left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right)^2 + \left(\frac{\sigma_x}{f_y/\gamma_{mo}}\right)\left(\frac{\sigma_z}{f_y/\gamma_{mo}}\right) + 3\left(\frac{\tau}{f_y/\gamma_{mo}}\right)^2 =$ — EN-1993-1-1 6.2.1(5)

Top Deck :	$\frac{-21.6^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.0^2}{fy/\gamma_{mo}}$	= $\frac{21.9}{318.2}$	= 0.07	≤ 1.00	OK	0.07 : U.R
Top Rib Web :	$\frac{-18.5^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{17.3^2}{fy/\gamma_{mo}}$	= $\frac{35.1}{318.2}$	= 0.11	≤ 1.00	OK	0.11 : U.R
Bott. Rib Web :	$\frac{-45.0^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{17.3^2}{fy/\gamma_{mo}}$	= $\frac{54.1}{318.2}$	= 0.17	≤ 1.00	OK	0.17 : U.R
Bott. Rib Flange	$\frac{-47.4^2}{fy/\gamma_{mo}}$	+	$\frac{0.0^2}{fy/\gamma_{mo}}$	-	$\frac{0.0}{fy/\gamma_{mo}}$	+ 3x	$\frac{2.6^2}{fy/\gamma_{mo}}$	= $\frac{47.6}{318.2}$	= 0.15	≤ 1.00	OK	0.15 : U.R

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Fatigue : 0.56 : U.R

Input Data :

γ_{mft} = 1.15
γ_{fft} = 1
N_{sc} = 10 $x10^6$ cycle
Impact Magnification factor : 1.000

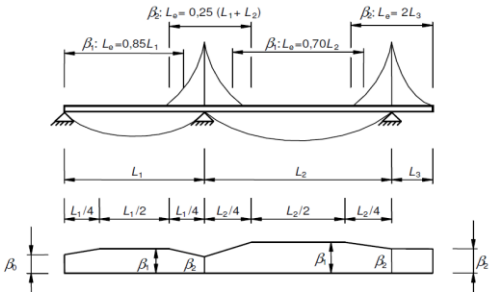
Straining actions :

M_x (+)= 46.57 KN.m
Axial= 2.072 KN
V_y = 81.54 KN
T = 0.791 KN.m

Effective Section Properties :

Shear lag Flanges :

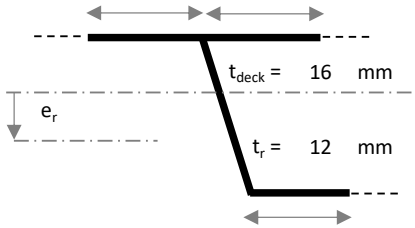
Sagging $b_{eff} = Bb_{c\text{-}eff}$



Effective Section Properties for sagging :
due to reduction of shear lag

$b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm $b_{eff} = Bb_{c\text{-}eff}$ = 153.5 mm
b_{red} = 6.545 mm b_{red} = 6.545 mm

$A_{red\text{-}up}$ = 418.9 mm^2 $y_{red\text{-}olc}$ 113.5 mm
$A_{red\text{-}bot}$ = 52.17 mm^2 $y_{red\text{-}olc}$ 248.5 mm
A_{new} = 20563.6 mm^2
$e_r = (A_{red}\, y_{old})/A_{new}$ = 1.681 mm



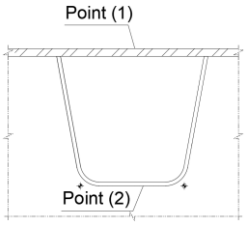
$I_{new} = I_{old} - (I_{red} + A_{new}\, e_r^2)$ = 392580543 mm^4

s_{top} = 3187672 mm^3 $y_{red.cg}$ = 123.2 mm $b_{eff} = Bb_{c\text{-}eff}$ = 107.8 mm
s_{bot} = 1552658.3 mm^3 b_{red} = 2.174 mm

Shear:
A_v = 8723 mm^2

Torsion:
A_o = 93396 mm^2

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Point (1) Upper deck Plate Gross Section

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y_{top} = 123.2 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 14.71 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 89.33 MPa 0.16 : U.R

fv deck= T / (2Ao tdeck) = 0.265 MPa < $\mu_r\, \tau_f / \gamma_{mft} =$ 56.78 MPa 0.00 : U.R

Point (2) Bottom Flange Plate Gross Section

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} =$ 1.00 $Y_{bot.}$ = 252.8 mm

f_{fn} = 118 MPa Table (19-a) i-(1) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 102.7 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 65.29 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 30.1 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 89.33 MPa 0.34 : U.R

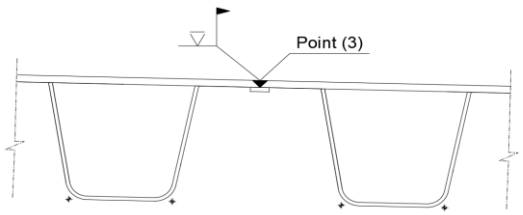
fv x (torsion)= 0.353 MPa < $\mu_r\, \tau_f / \gamma_{mft} =$ 56.78 MPa 0.01 : U.R

Point (3) Deck weld Butt weld one side (longitudinal)

$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 123.2 mm

f_{fn} = 83 MPa Table (19-b) ii-(10&11) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 72.26 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 14.71 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 62.83 MPa 0.23 : U.R



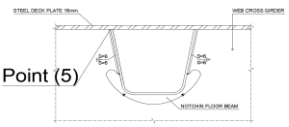
Point (4) web at cutout (local effects) Rib web at cutout

Not applicable

0.00 : U.R

0.00 : U.R

Point (5) web weld (longitudinal stresses) Butt & Fillet weld two side (longitudinal)



$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} =$ 1.00 Y = 107.2 mm

f_{fn} = 92 MPa Table (19-b) i-(9) $f_f = f_{fn}\sqrt[3]{\frac{5x\,10^6}{N_{sc}}}$ = 73.02 MPa

τ_{fn} = 75 MPa $\tau_f = \tau_{fn}$ = 65.22 MPa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 12.81 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 63.5 MPa 0.20 : U.R

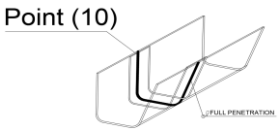
fv y= 9.348 MPa fv y (torsion)= 0.353 MPa

$f_{v\,y(total)}$ = 9.70 MPa < $\mu_r\, \tau_f / \gamma_{mft}$ = 56.7 MPa 0.17 : U.R

Point (10) Welded splice Welded rib splice

Straining actions of Welded splice at location of 0.7m from F.B

Mx =	N
37.4	15
kN.m	kN



$\mu_r = \left(\frac{25}{t_r}\right)^{0.25}$ =1.000 $Y_{bot.}$ = 252.8 mm

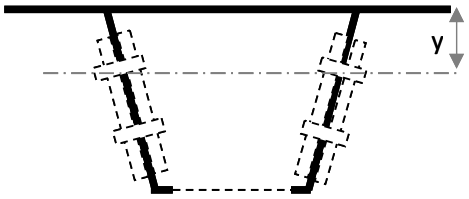
f_{fn} = 80 MPa Euro Table (8-8) (4) N=2x10^6 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}}$ = 51.31 MPa

f_{fn} = 58.94 MPa N=5x10^6

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A}$ = 24.82 MPa < $\mu_r\, f_f / \gamma_{mft}$ = 44.62 MPa 0.56 : U.R

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Section 2-2 (Splice): Straining action of fatigue at Splice



A =	17637	mm^2	$S_{x\text{-top}}$ =	3031147	mm^3
y =	95	mm	$S_{x\text{-bot}}$ =	1027924	mm^4
I_x =	288621959	mm^4			

web

No.of rows	3
Hole Diameter	30

Point (6) Upper deck Plate Butt weld one side (Transverse)

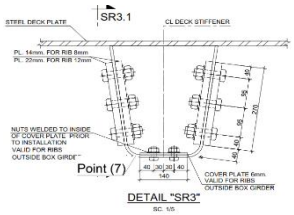
$$\mu_r = \left(\frac{25}{t_d}\right)^{0.25} = 1.000 \qquad Y_{yp.} = 95.2 \text{ mm}$$

f_{fn} = 52 MPa Table (19-b) ix-(23) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 MPa

$$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 15.5 \text{ MPa} \quad < \quad \mu_r\, f_f / \gamma_{mft} = 39.4 \text{ MPa}$$

0.39 : U.R

Point (7) Bottom Flange Plate Net Section (Bollted)

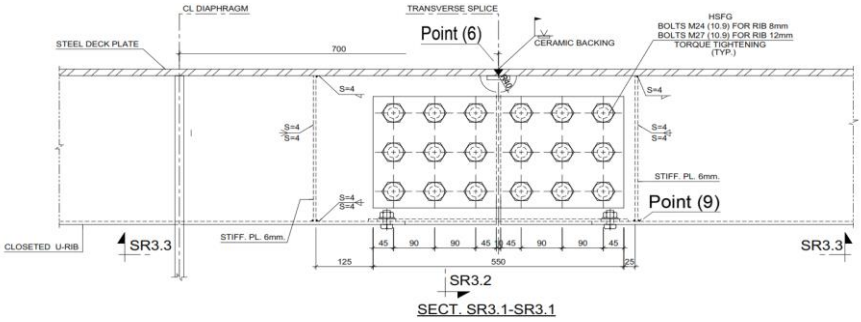


$$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000 \qquad Y_{bot.} = 281 \text{ mm}$$

f_{fn} = 103 MPa Table (19-a) ii-(5) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 89.7 MPa

$$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} = 45.4 \text{ MPa} \quad < \quad \mu_r\, f_f / \gamma_{mft} = 78 \text{ MPa}$$

0.58 : U.R



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Point (9) Web Vertical stiffeners Vertical stiffeners Fillet welded

for sagging

Straining actions and gross section

Mx =	N
0	0

Bolted splice at location of 0.7m and closure plate at 1.0m from F.B

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.}$ = 240.8 mm

f_{fn} = 59 MPa Table (19-b) xviii-(36&37 $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 51.4 MPa

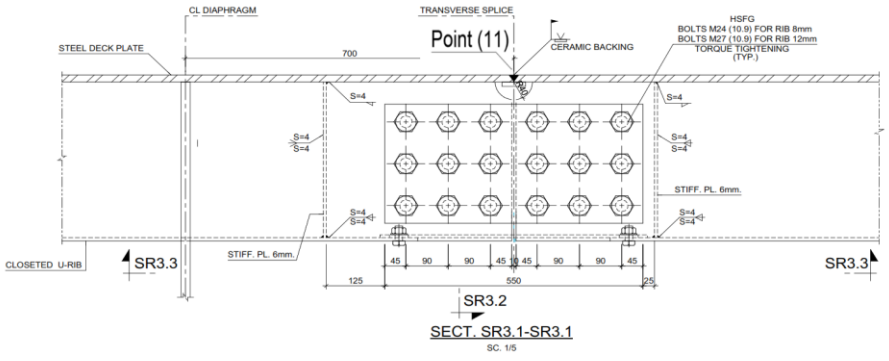
$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 0 MPa < $\mu_r\, f_f / \gamma_{mft} =$ 44.7 MPa 0.00 : U.R

Point (11) Intermittient Long. Welc Zones containing cope holes

$\mu_r = \left(\frac{25}{t_r}\right)^{0.25} = 1.000$ $Y_{bot.}$ = 79.2 mm

f_{fn} = 52 Mpa Table (19-b) v-(15) $f_f = f_{fn}\sqrt[5]{\frac{5x\,10^6}{N_{sc}}} =$ 45.3 Mpa

$\gamma_{fft} f_{act} = \frac{M\,Y}{I} + \frac{N}{A} =$ 12.9 Mpa < $\mu_r\, f_f / \gamma_{mft} =$ 39.4 Mpa 0.33 : U.R



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Plate.G : 0.99 : U.R

Bolts : 0.84 : U.R

Splice Straining actions :

M = 77.83 KN.m

Q = 127.4 KN

P = 42.95 KN

T = 5.934 KN.m

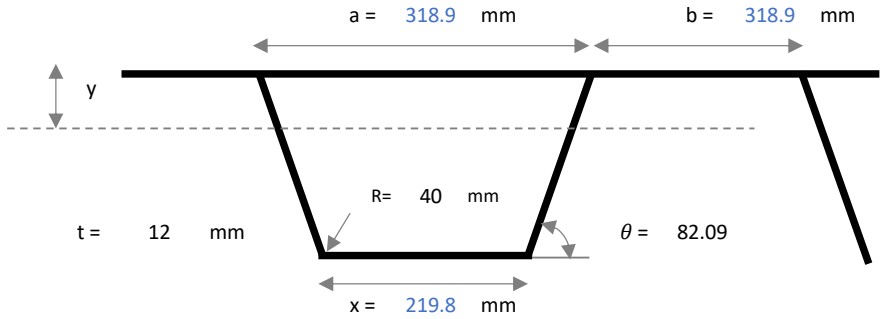
AVERAGE Global Rib stress = -55.26 Mpa

Eq. Global Normal Force = -1160 kN

$N_{Rib\text{-}total}$ = 1203 KN

$F_{y\text{-}actual}$ 350 Mpa

Section Properties :



A = 20997 mm^2

y = 122 mm

I_x = 400631140 mm^4

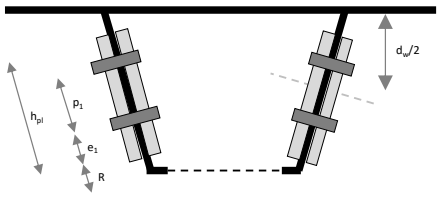
$S_{x\text{-}top}$ = 3293897 mm^3

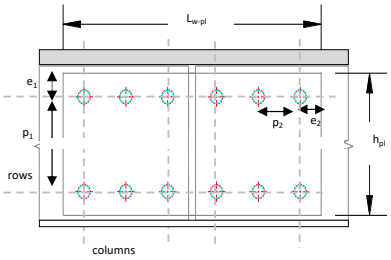
$S_{x\text{-}bot}$ = 1574983 mm^4

$I_{x\text{-}rib}$ = 268650721 mm^4 0.67

Splice plates Dimintions & BOLTES Confiugrations :

Bolts Grade : 10.9





web :

Plates : h_{pl} = 270 mm

< 271 mm (233)

$L_{w\text{-}pl}$ = 532 mm

t_{pl} = 24 mm > 20 mm

> 40 mm

$y_{splice\text{-}web}$ = -13.34 mm $F_{y\ pl.}$ = 330 Mpa

Bolts : M27

No.of Rows 3 $y_{bot\text{-}pl.edge}$ = 43 mm

No.of column 3 $b_{opening}$ 100 mm

e_1 = 41 mm ≥ 1.5 Ø = 40.5 mm

p_1 = 94 mm ≥ 3.0 Ø = 81 mm

e_2 = 40.5 mm ≥ 1.5 Ø = 40.5mm

p_2 = 90 mm ≥ 3.0 Ø = 81 mm

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tension
compression

Check on base material :

(Befor opening net section) :

						$y_{bot-pl.edge}$	y_{Bolt-1}	y_{Bolt-2}	y_{Bolt-3}
A =	18837	mm^2	S_{x-top} =	3313146	mm^3	43	84	177	270
y =	113	mm	S_{x-bot} =	1418548	mm^4				
I_x =	373470022	mm^4							

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1203}{19}-\frac{77.83}{3.313}=$ 43 − 16 = 26.9 < $.9f_u/1.25$ 352.8 OK U.R : 0.076

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1203}{19}-\frac{77.83}{1.419}=$ 43 + 37 = 79.2 < $.9f_u/1.25$ 352.8 OK U.R : 0.224

$0.9\,A_n F_u/\,1.25$ =	6646	KN	<	$A_g F_y/\,1.1$ =	6681	KN
$0.9\,Sx_n F_u/\,1.25$ =	500	KN.m	<	$Sx_g F_y/\,1.1$ =	501	KN.m

(After one bolt column at opening section without bolt holes) :

With 0.6667 Force clause-509.6.3.3

						$b_{opening}$
A_{total} =	19797	mm^2	0.94	S_{x-top} =	3022371	mm^3	100
y =	107	mm		S_{x-bot} =	1195513	mm^4	
Ix =	322103396	mm^4		t_{web} =	12	mm	

clause 609.6.1.2

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1203}{20}-\frac{77.83}{3.022}=$ 41 − 17 = 23.3 < $f_y/1.1$ = 318.2 OK U.R : 0.073

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)=$ $\frac{1203}{20}-\frac{77.83}{1.196}=$ 41 + 43 = 83.9 < $f_y/1.1$ = 318.2 OK U.R : 0.264

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(After one bolt column at opening net section with bolt holes) : With 0.6667 Force

A_{total} = 17637 mm^2 $S_{x\text{-}top}$ = 3031147 mm^3 clause-509.6.3.3

y = 95 mm $S_{x\text{-}bot}$ = 1027924 mm^4

Ix = 288621959 mm^4

-Top stresses: (compression)

$0.6667\left(\frac{N}{A}+\frac{M_x}{S_{dx}}\right)$ = 1203/18 + 77.83/3.031 = 45 − 17 = 28.4 < .9f_u/1.25 352.8 OK U.R : 0.08

-Bottom stresses: (tension)

$0.6667\left(\frac{N}{A_d}+\frac{M_x}{S_{dx}}\right)$ = 1203/18 + 77.83/1.028 = 45 + 50 = 96.0 < .9f_u/1.25 352.8 OK U.R : 0.272

WEB :

Actual Force = 150.47 KN 0.13 : U.R

Yielding	<	V_d = Awg Fy /1.1√3 =	1410.8	KN	clause-506.1.2
Shear Rupture	<	V_{dr} = 0.9 Awn Fu /1.25√3 =	1124.4	KN	
Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	4380.7	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	5126.3	KN	

A_{vn} = 8634.1 mm^2 A_{tn} = 6984 mm^2

A_{vg} = 10434 mm^2 A_{tg} = 10584 mm^2

Actual tension Force A_{rib}/A_{tot} = 595.4 KN 0.21 : U.R

Block shear	<	T_{db} =(0.9 A_{tn} F_u /1.25)+(A_{vg} F_y /1.1√3) =	2917.0	KN	clause-506.1.3
	<	T_{db} =(A_{tg} F_y /1.1)+(0.9 A_{vn} F_u /1.25√3) =	2776.2	KN	

for Rib : A_{vn} = 6984 mm^2 A_{tn} = 3072 mm^2 t= 12 mm

A_{vg} = 10584 mm^2 A_{tg} = 4512 mm^2

for Splice Plate : A_{vn} = 17568 mm^2 A_{tn} = 12288 mm^2 $2xt_{pl}$ = 48 mm

A_{vg} = 21168 mm^2 A_{tg} = 18048 mm^2

Elliptical BSR :

$\left(\frac{V}{\phi R_{bsv}}\right)^2+\left(\frac{H}{\phi R_{bsh}}\right)^2$ = 0.0343 + 0.2145 = 0.24 < 1 0.24 : U.R

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Check on Splice plates :

rib :

Area web splice Agross = 25920 mm^2 ≥ 1.05 Area rib = 10910.30314 mm^2 clause-509.6.3.3 — 0.42 : U.R

Section :

Ix splice = 422800931.1 mm^4 ≥ 1.05 Ix Section = 420662697 mm^2 — 0.99 : U.R

Check on Splice Bolts :

WEB :

$M_{x\text{-web}}$ =	26.094	KN.m	No.of Rows	3	p1 =	94	mm
$M_{due\ to\ shear}$ =	5.7886	KN.m	No.of column	3	p2 =	90	mm
$N_{one\text{-}web}$ =	297.7	KN	no.of Bolts =	9	$X_{C.G}$ =	90	mm
$V_{Torsion} = T/b_{flange}$ =	23.059	KN			$Y_{C.G}$ =	94	mm
$V_{shear} + V_{Torsion}$ =	86.766	KN					
0.5 $M_{x\text{-capacity-rib}}$ =	168.02	KN.m					

$Q_{v\text{-}y}$ =	9.6407	KN	$Q_{T\text{-}y}$ =	28.238	KN	Q_y =	37.878	KN
$Q_{N\text{-}x}$ =	33.078	KN	$Q_{T\text{-}x}$ =	29.493	KN	Q_x =	62.571	KN

1.05 Q_{total} = 76.8 KN one bolt < $V_{nsf}/\gamma = \mu$ n K F/γ = 257.04 KN μ = 0.5 — v= 76.07

0.5 capacity-rib = 215.18 KN < — t= 10.57

F_o =	$A_{nb}\ f_o$ =	321.3	KN	n =	2
f_o =	700	N/mm^2		K_h =	1
A_{nb} =	459	mm^2		γ =	1.25

0.84 : U.R